



Department of Computer Science

COMP4300 - Graduation Project

Spring 2025/2026



Navigo وصلني

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June 28, 2026

Section – B

Title of Project: Navigo – وصلني

Project No: 14

Supervisor: Mr. Fadi Khalil

Key Areas: Mobile Application Development, Real-Time Tracking, GPS/Maps, Cloud Backend, Smart Transportation

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Abstract

Navigo – وصالني is a mobile application designed to improve the organization and reliability of daily public transportation in Palestine. The project addresses the absence of a unified digital platform that supports licensed public transportation services such as minibuses, taxis, and buses operating on local and university routes.

The implemented system connects passengers with licensed drivers through a simple and user-friendly mobile application. It enables passengers to view nearby vehicles, check seat availability, track real-time vehicle locations, and receive accurate arrival time estimates. Drivers can manage trip status, update availability, and respond efficiently to passenger requests. In addition, an administrative panel is provided to manage drivers, routes, and daily transportation operations.

By combining real-time tracking, cloud-based services, and an intuitive interface, Navigo – وصالني provides better transportation organization, safety, and predictability. The system helps reduce waiting times and improves the commuting experience for university students and the wider community.

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Acknowledgements

We want to express our sincere gratitude to everyone who supported us throughout this project. First, we would like to sincerely thank our supervisor, Mr. Fadi Khalil, for his continuous guidance, valuable feedback, and constructive suggestions; his expertise and encouragement played a major role in improving our work and keeping us on the right path from start to finish. We also extend our appreciation to the Computer Science Department for providing a strong academic environment, helpful resources, and ongoing support that contributed to developing our knowledge and skills and made this project possible. In addition, we are deeply grateful to our families for their patience, understanding, and unconditional support throughout the demanding stages of this work; their motivation and trust helped us stay focused and overcome challenges. Finally, we would like to thank our friends for their help, cooperation, and moral support, and for making the journey more enjoyable through discussion, teamwork, and shared effort.

Chapter 1

Introduction

1.1 General Overview

For many students and daily commuters in Palestine, navigating public transportation can be a frustrating and time-consuming ordeal. Waiting for buses or micro buses often involves uncertainty, unpredictability, and wasted time. Public transportation is a lifeline for residents, especially in high-demand corridors such as Ramallah and major university routes, where access to private vehicles is limited. Yet, despite its essential role, the current system remains largely manual and unstructured. Passengers frequently face a lack of reliable information regarding vehicle arrival times, seat availability, and route updates, which not only heightens stress but also diminishes overall commuting efficiency.

In parallel, drivers face similar operational challenges. Without digital coordination tools, it becomes difficult to manage passenger interactions, organize trips, and maintain consistent schedules. These limitations contribute to extended waiting times, low service predictability, and an overall commuting experience that does not scale well with growing demand. As the number of students and daily commuters continues to rise, the need for a dependable, technology-enabled transportation solution becomes increasingly urgent.

This project presents the analysis, design, implementation, and testing of a smart, mobile-based transportation system aimed at introducing structure, transparency, and effective communication into the public transport process. The implemented system modernizes traditional transportation practices by enabling real-time vehicle tracking, facilitating communication between passengers, drivers, and route managers. By leveraging mobile technologies and cloud-based services, the solution provides a scalable foundation intended to enhance reliability and predictability for the local community.

1.2 Motivation

For many people in Palestine, using public transportation is a daily source of frustration rather than convenience. Passengers often spend long periods waiting without knowing when a bus or microbus will arrive, or whether seats are available. This uncertainty leads to delays, unsafe waiting conditions, and unnecessary stress, particularly for students and daily commuters who depend on public transportation to reach universities, workplaces, and city centers.

Drivers face similar difficulties, as the lack of structured communication and coordination tools makes it challenging to organize trips and respond efficiently to passenger demand. The absence of a unified digital system that provides real-time vehicle tracking, seat availability, and accurate arrival time estimates further contributes to inefficiency for both passengers and drivers. These challenges highlight the urgent need for a practical, technology-driven solution that improves reliability and restores confidence in public transportation.

1.3 Method

The Navigo-*وصلي* project followed an iterative development approach that included requirements analysis, system design, implementation, and functional testing. The work started by studying the transportation problem, identifying user needs, and defining the main requirements for passengers, drivers, route managers, and administrators.

After the requirements were identified, the system analysis and design phase was completed. This phase included defining the system architecture, preparing UML diagrams, designing the data model, and organizing the main modules of the application. The design phase helped guide the implementation and ensured that the system features were clearly connected to the project requirements.

The implementation phase was then carried out using Flutter for the mobile application interfaces and Firebase for backend services. Firebase Authentication and OTP verification were used for user authentication, while Firebase database services and security rules supported data storage, synchronization, and role-based access control. The implemented modules include the Passenger module, Driver module, Route Manager module, and Admin panel.

Core transportation features were implemented as part of the system, including route browsing, trip scheduling, trip requests, real-time tracking, notifications, and cash payment handling. The system maintains and updates seat availability automatically based on passenger reservations and the capacity assigned to each trip.

Finally, the system was validated through functional testing. The tests covered registration, OTP verification, route browsing, trip scheduling, seat availability calculation, trip requests, live tracking, notifications, support reports, and administrative workflows. This methodology helped move the project from analysis and design into a working ap-

plication that was evaluated against its main functional requirements.

1.4 Aim and Objectives

1.4.1 Aims

The main aim of this project is to design and implement a smart and organized public transportation system that enhances communication between passengers and drivers. The system was also tested to verify its main functions. It addresses current transportation challenges by providing a structured, predictable, and technology-supported solution suitable for students and residents.

1.4.2 Objectives

The objectives of this project are as follows:

- Analyze the current public transportation process and identify key challenges faced by passengers and drivers.
- Create a mobile application that connects passengers with nearby drivers operating on fixed or semi-fixed routes.
- Provide real-time vehicle tracking and accurate arrival time estimation (ETA).
- Display seat availability information to support informed passenger decisions.
- Allow passengers to browse routes, view route details, and request trips easily.
- Enable route managers to create and manage trips, while allowing drivers to update trip status and handle passenger requests.
- Provide an administrative panel to manage drivers, routes, and overall system activity.
- Improve organization, communication, and efficiency in public transportation through the implemented system.

1.5 Overview of the technical area

The Navigo-وصّلي system was developed using a set of modern technologies and tools that support real-time data exchange, mobile accessibility, and scalable system architecture. The following technologies supported the design and implementation of the system.

1.5.1 Backend and Database: Firebase

Firebase is used as the backend platform to support real-time data synchronization, user authentication, and cloud-based data storage. Its real-time database capabilities make it suitable for handling dynamic transportation data such as trip status, vehicle location, and seat availability [1].

1.5.2 Mobile Application Framework: Flutter

Flutter was selected as the mobile development framework due to its cross-platform capabilities, allowing a single codebase to support both Android and iOS applications. Flutter supports rapid interface development and a consistent user experience across different devices [2].

1.5.3 Map and Location Services: Google Maps API

Google Maps Platform APIs support map visualization, route display, location search, and real-time location services within the application. The integration of mapping services enables vehicle tracking, route representation, and distance estimation as part of the system design [3].

1.5.4 Design and Prototyping Tool: Figma

Figma is used during the design phase to create wireframes and user interface prototypes. These prototypes help visualize user interactions and ensure consistency in the system's interface design before implementation [4].

1.5.5 Version Control System: GitHub

GitHub was used as the version control platform to support collaborative development, source code management, and version tracking during implementation [5].

1.5.6 Future Payment Services: Jawwal Pay

Digital payment services such as Jawwal Pay are excluded from the current system requirements and are considered as a future enhancement. In a later version, these services may support secure cashless fare payment options and improve transaction convenience for users [6].

Overall, the selection of these technologies provided a scalable and flexible technical foundation for the implemented Navigo-وصّلي system.

1.6 Overview of the report

This report is organized into several chapters that present the complete development of the Navigo-وصلي system. Chapter 2 provides the project background and related work. Chapter 3 presents the system analysis and design, including requirements, models, architecture, and data management. Chapter 4 describes the implementation of the passenger, driver, route manager, and administrator modules. Chapter 5 presents the testing and evaluation of the implemented system. Chapter 6 concludes the report and discusses system limitations and future work.

The implemented system is particularly relevant to university routes such as Birzeit–Ramallah, a daily challenge for thousands of students relying on public transportation.

Chapter 2

Background

2.1 Details of Relevant Theory

Smart transportation systems have emerged as an important research area aimed at improving the efficiency, safety, and reliability of public transportation services. These systems rely on the integration of information and communication technologies to support data-driven decision making, real-time monitoring, and enhanced coordination between transportation stakeholders [7].

One of the core components of smart transportation systems is the use of mobile applications. Mobile platforms enable direct interaction between users and transportation services by providing access to real-time information such as vehicle location, arrival time estimates, and service availability. The widespread adoption of smartphones has made mobile-based transportation solutions a practical and scalable approach for improving daily commuting experiences.

Real-time systems play a critical role in transportation applications, particularly in scenarios that require continuous updates and timely information delivery. Technologies such as Global Positioning Systems (GPS) allow vehicles to be tracked accurately, while real-time data processing enables dynamic updates of trip status and arrival predictions. These capabilities help reduce uncertainty and improve predictability for passengers and drivers alike [8].

Cloud computing further supports smart transportation systems by providing scalable infrastructure for data storage, processing, and communication. Cloud-based platforms allow transportation systems to handle large volumes of dynamic data while maintaining availability and performance. In addition, cloud services facilitate integration with external systems and support future system expansion without major infrastructure changes [9].

Together, mobile technologies, real-time data processing, and cloud computing form the theoretical foundation of modern smart transportation systems. These concepts provide the basis for designing transportation platforms that are responsive, scalable, and capable of addressing the challenges of contemporary urban mobility.

2.2 Review of Past / Reported Work

Smart transportation and mobility applications have been widely adopted worldwide to improve the organization, accessibility, and efficiency of public transportation systems. Many existing solutions focus on route planning, real-time arrival information, and trip management across different transportation modes such as buses, metro systems, taxis, and ride-hailing services. The following applications represent notable examples of such systems.

2.2.1 S'hail (Dubai Smart Transport Application)

S'hail is a comprehensive smart mobility application developed by Dubai's Roads and Transport Authority (RTA). It integrates multiple transportation modes, including Metro, Tram, Bus, Marine, and Taxi services, as well as third-party platforms such as Uber and Careem. The application allows users to plan trips, view real-time schedules, track vehicles using GPS, and locate nearby stations. In addition, it supports digital payments and service management features, providing an integrated transportation platform [10].

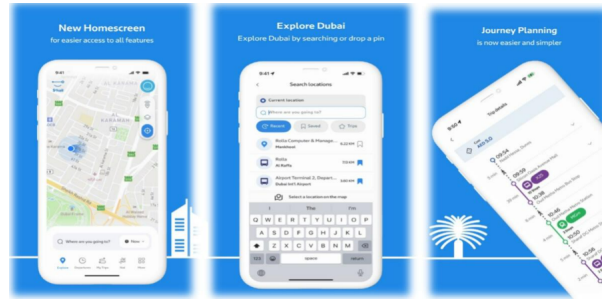


Figure 2.1: S'hail Application

2.2.2 Citymapper

Citymapper is a global navigation and trip-planning application used in major cities such as London, New York, and Paris. It provides multimodal route suggestions, real-time arrival information, and turn-by-turn navigation across buses, metro systems, trains, and walking routes. Although Citymapper does not support seat booking or direct communication with drivers, it demonstrates effective route visualization and transport data presentation, which are relevant to modern transportation systems [11].

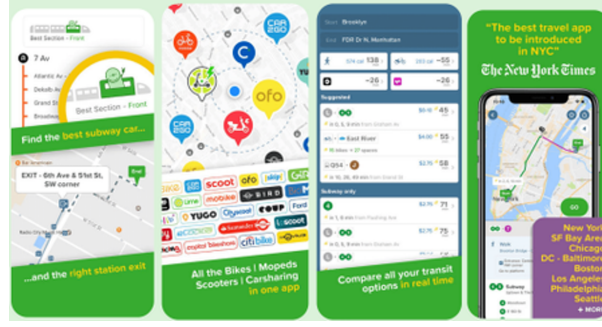


Figure 2.2: Citymapper Application

2.2.3 Careem

Careem is a widely used ride-hailing application operating in the Middle East, North Africa, and South Asia. It connects passengers with nearby drivers for on-demand and scheduled trips. The application provides features such as GPS-based tracking, estimated arrival times, in-app payments, ride history, and safety tools. Careem also includes management features that allow administrators to monitor drivers and trip activity, highlighting structured coordination between users and service providers [12].

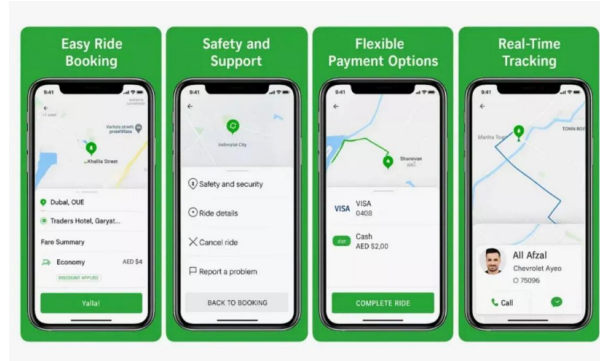


Figure 2.3: Careem Application

2.2.4 JETT application

The JETT mobile application is designed to provide passengers with a convenient, reliable, and user-friendly platform for managing their travel experience. Through the app, users can easily search for available trips, view schedules, book tickets, select seats, and receive real-time notifications about trip updates. The application aims to reduce waiting times, improve accessibility to transportation services, and enhance overall passenger satisfaction by offering secure payments, digital tickets, and clear travel information in one integrated system [13].

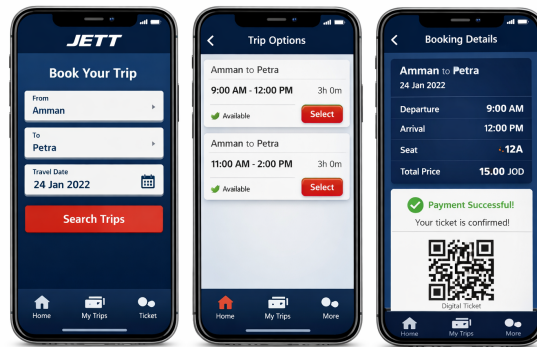


Figure 2.4: JETT Application

2.2.5 Moovit

Moovit is a global public transportation application operating in over 100 countries. It aggregates data from buses, trains, metro systems, shared rides, and walking routes. Moovit provides real-time tracking, optimized route planning, delay alerts, and step-by-step navigation. Its ability to integrate multiple transportation modes aligns with the objectives of improving efficiency and accessibility in urban transportation systems [14].

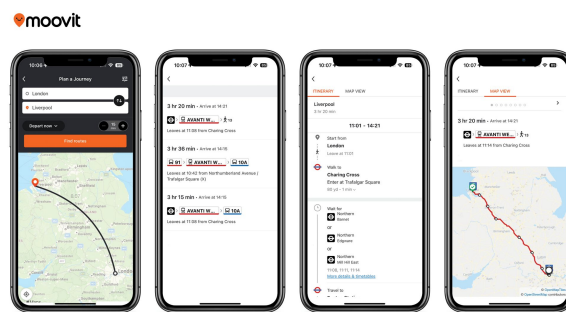


Figure 2.5: Moovit Application

Although existing transportation applications offer features such as navigation and real-time tracking, most of them are designed for ride-hailing services or large metropolitan transit systems. Applications like Careem focus on private, on-demand trips, while Citymapper and Moovit provide route information without supporting seat availability, booking, or passenger-driver coordination. In contrast, Navigo-وصلني is specifically designed for licensed public transportation within the Palestinian context, focusing on fixed routes, real-time seat availability, and local administrative control for university and city commuting.

2.2.6 Comparison Summary

Feature	Navigo	JETT	S'hail	Citymapper	Careem	Moovit
Licensed Public Transport	✓	✓	×	×	×	×
Fixed / Route-Based Lines	✓	✓	✓	×	×	×
Real-Time Vehicle Tracking	✓	×	✓	✓	✓	✓
Seat Availability / Seat Selection	✓	✓	×	×	×	×
Trip Booking (Public Transport)	✓	✓	×	×	×	×
Digital Payments	Future	✓	✓	×	✓	×
Route Optimization	✓	×	✓	✓	✓	✓
Passenger Notifications	✓	✓	✓	✓	✓	✓
Multimodal Support	✓	×	✓	✓	×	✓
Designed for Local Palestinian Context	✓	×	×	×	×	×

Table 2.1: Comparison between Navigo-وصلني and existing transportation applications

2.3 Tools and Technology

The development of Navigo-وصلتي relies on a set of modern tools and technologies that support real-time tracking, intelligent route management, secure communication, and cross-platform mobile application development. These technologies form the technical foundation of the implemented system and support scalability, reliability, and usability for passengers, drivers, and administrators.

2.3.1 Backend and Database: Firebase

Firebase is used as the primary backend platform for the system. It manages real-time data related to trips, drivers, passengers, and seat availability. Firebase provides continuous data synchronization between users and drivers, secure authentication mechanisms, and support for push notifications. These features enable instant updates for ride requests, trip status changes, and arrival alerts, which are essential for real-time transportation applications [1].

2.3.2 Mobile Development Framework: Flutter

Flutter is used to develop the mobile application for both Android and iOS platforms using a single shared codebase. It offers high performance, a flexible user interface framework, and rapid development through features such as Hot Reload. These capabilities make Flutter suitable for implementing dynamic and interactive transportation features while maintaining consistency across platforms [2].

2.3.3 Map and Location Services: Google Maps API

Google Maps API is integrated into the system to provide route visualization, vehicle location tracking, distance estimation, and navigation support. These services enable real-time monitoring of vehicle movement and improve passenger awareness during trip planning and execution. Accurate location services are a core requirement for intelligent transportation systems [3].

2.3.4 Design and Prototyping Tool: Figma

Figma is used during the design phase to create wireframes and interactive prototypes for the system. It supports collaborative interface design and allows the development team to evaluate user workflows before implementation. Using Figma helps ensure a consistent, clear, and user-friendly interface for all system roles [4].

2.3.5 Version Control and Collaboration: GitHub

GitHub is used for version control and source code management throughout the project. It supports collaborative development, change tracking, and organized workflow management. GitHub also provides secure cloud-based storage for the project repository, ensuring code integrity and ease of collaboration among team members [5].

Chapter 3

System Analysis and Design

This chapter presents the system analysis and design of the Navigo-وصلني transportation platform. It describes the system model, architecture, functional components, and design representations used to guide the development of the implemented system.

The chapter begins with a detailed product description, including system objectives, main features, operating environment, constraints, and requirements. It then introduces the functional decomposition of the system using use case analysis, followed by system models such as class, sequence, activity, and state chart diagrams. Finally, the system architecture and data management design are presented to illustrate how system components interact and how data is stored and managed.

3.1 System Model and Architecture

This section presents the system model and architectural overview of the Navigo-وصلني transportation platform. It provides a comprehensive description of the system, including its objectives, main features, operating environment, constraints, and system requirements.

3.1.1 Product Description

Navigo-وصلني is a smart mobile-based transportation system designed to improve the organization and reliability of licensed public transportation in Palestine. The system focuses on serving university students and city commuters who depend on minibuses, taxis, and buses operating on fixed or semi-fixed routes.

The implemented system connects passengers with nearby licensed drivers through an interactive mobile application. It allows passengers to view available vehicles on a map, check seat availability, track driver locations in real time, and estimate arrival times accurately. Drivers can manage trip status, update availability, and receive trip requests from passengers. In addition, the system includes administrative functionalities that enable route and driver management, ensuring better control and monitoring of transportation activities.

Navigo-وصلني is designed as a scalable platform that can be extended to support additional routes, users, and services in the future.

System Objectives

The main objectives of the Navigo-وَصْلَني system are as follows:

- To provide an organized and technology-supported solution for public transportation.
- To reduce passenger waiting time by offering real-time vehicle tracking and arrival estimates.
- To improve communication and coordination between passengers and drivers.
- To enhance transportation safety by working only with verified and approved drivers.
- To support efficient route and trip management through administrative control.
- To provide a reliable and predictable commuting experience for students and residents.

System Main Features

The system includes the following main features:

- **User Authentication:** Secure registration and login for passengers, drivers, and administrators.
- **Real-Time Vehicle Tracking:** Live GPS-based tracking of vehicles on an interactive map.
- **Seat Availability Display:** Real-time information about available seats in each vehicle.
- **Trip Request Management:** Passengers can request trips and track driver arrival.
- **Trip Management:** Route Managers create and manage trips, while drivers update trip status and handle assigned passengers.
- **Administrative Control:** Admins can approve drivers, manage routes, monitor system activity, and review statistics.
- **Notifications and Alerts:** Real-time notifications for trip updates, driver arrival, and system announcements.
- **Cash Payment Support:** Trip fares are handled through cash payment collected by the driver.

Operating Environments

Navigo-وصلني is designed to operate within a mobile and cloud-based environment. The system is intended to run on the following environments:

- **Mobile Devices:** Android and iOS smartphones used by passengers and drivers.
- **Cloud Backend Environment:** Firebase cloud services for real-time data storage, synchronization, and authentication.
- **Internet Connectivity:** Continuous internet access is required to support real-time tracking, notifications, and data updates.
- **Map and Location Services:** Integration with Google Maps API to support GPS tracking, route visualization, and location-based services.

The system is designed to support urban and semi-urban environments, particularly university and city transportation routes.

Constraints

The design of the Navigo-وصلني system is subject to several constraints, including:

- **Internet Dependency:** The system requires a stable internet connection to support real-time features such as vehicle tracking and notifications.
- **GPS Accuracy:** Vehicle tracking accuracy depends on the availability and quality of GPS signals and device sensors.
- **Regulatory Constraints:** The system is limited to licensed public transportation vehicles and approved drivers only.
- **Platform Limitations:** The system is designed for mobile platforms and may not fully support older devices.
- **Project Scope:** The project covers analysis, design, implementation, and functional testing within the confirmed academic project scope.

Functional User Requirements

The following user requirements define the expected functionalities of the Navigo-وصلني system from the perspective of passengers, drivers, and administrators:

- **UR1.0 Passenger and Driver Registration:** Passengers and drivers shall be able to create an account by entering personal information. Different required fields shall be provided for passengers and drivers.
- **UR2.0 Route Browsing and Filtering:** Users shall be able to search for routes by selecting a start point and destination, then filter available trips using Now or Scheduled travel options.
- **UR3.0 Trip Request:** Users shall be able to submit a trip request by selecting a pickup location using GPS or manual input, then filtering available trips based on date, time, vehicle type, and route before confirming the request.

- **UR4.0 Real-Time Tracking:** The system shall allow users to track vehicles in real time. After viewing route and tracking information, users shall be able to submit a trip request, and the system shall assign an available driver or vehicle.
- **UR5.0 Trip Confirmation and History:** Users shall receive trip confirmation details and shall be able to view their past trip history.
- **UR6.0 Notifications and Live Updates:** Users shall receive real-time notifications regarding trip status, delays, and driver arrival.
- **UR7.0 Cash Payment Handling:** Users shall be able to complete trip payment using cash collected by the driver.
- **UR8.0 Route and Vehicle Management (Admin Panel):** Administrators shall be able to manage routes, pricing, vehicles, and driver accounts through an administrative panel.
- **UR9.0 Safety, Support, and Reporting:** Users shall have access to customer support features, safety options, and issue reporting mechanisms.
- **UR10.0 Map, GPS, and Navigation Services:** The system shall utilize map and GPS technologies to provide accurate location tracking and navigation services.

Functional System Requirements

The following system requirements define the detailed functional and technical behavior of the Navigo – *نavigo* system. Each set of system requirements corresponds to a specific user requirement.

UR1.0 Passenger and Driver Registration

- **SR1.1:** The system shall provide separate registration forms for passengers and drivers with required fields specific to each role.
- **SR1.2:** The system shall require passengers to provide full name ,and phone number during registration.
- **SR1.3:** The system shall require drivers to provide full name, phone number, vehicle type, plate number, driver license,and route line during registration.
- **SR1.4:** The system shall verify user phone numbers using One-Time Password (OTP).
- **SR1.5:** The system shall authenticate registered users via their phone number.
- **SR1.6:** The system shall allow users to update their profile information after registration.

UR2.0 Route Browsing and Filtering

- **SR2.1:** The system shall allow users to search for available routes by selecting a start point and an end point.
- **SR2.2:** The system shall allow users to filter the available trips for the selected route based on travel time options (Now or Scheduled).
- **SR2.3:** The system shall allow authorized administrators to add, edit, or remove route listings.

UR3.0 Trip Request

- **SR3.1:** The system shall allow users to submit a trip request by selecting a pickup location (GPS or manual), route, travel time, date and vehicle type.
- **SR3.2:** The system shall process the trip request and display available vehicles or drivers.
- **SR3.3:** The system shall allow users to cancel or modify a trip before it starts.

UR4.0 Real-Time Tracking

- **SR4.1:** The system shall allow the Route Manager to create trips and assign the appropriate driver and vehicle to each trip.
- **SR4.2:** The system shall display trip request details, including assigned driver and vehicle information.
- **SR4.3:** Users shall be able to track the driver's real-time location on the map.
- **SR4.4:** The system shall update driver status in real time (Available, On Trip, Offline).

UR5.0 Trip Confirmation and History

- **SR5.1:** The system shall send trip confirmation details via push notification.
- **SR5.2:** The system shall allow users to view trip details including driver name, vehicle type, route, and estimated fare.
- **SR5.3:** The system shall maintain a history log of past trips accessible to users.
- **SR5.4:** The system shall allow drivers to update trip status (arrived, started, completed), with updates reflected to users within 30 seconds.

UR6.0 Notifications and Live Updates

- **SR6.1:** The system shall notify users when drivers accept trips and when drivers are approaching the pickup location.
- **SR6.2:** The system shall notify users of delays or route changes with updated estimated arrival times.
- **SR6.3:** The system shall send notifications primarily via push notifications, with SMS fallback for verification or when push notifications are unavailable.

UR7.0 Cash Payment Handling

- **SR7.1:** The system shall support cash payment for trip fares.
- **SR7.2:** The system shall display the estimated fare before trip confirmation.
- **SR7.3:** The system shall mark the trip payment status as cash pending until collection.
- **SR7.4:** The driver shall be able to confirm that cash payment has been collected.

UR8.0 Route and Vehicle Management (Admin Panel)

- **SR8.1:** Administrators shall be able to add, edit, or delete routes and update pricing.
- **SR8.2:** Administrators shall manage vehicle availability and categories.
- **SR8.3:** Administrators shall activate or deactivate driver accounts.
- **SR8.4:** The system shall enforce role-based access control for Admin, Driver, and Passenger users.

UR9.0 Safety, Support, and Reporting

- **SR9.1:** The system shall allow users to submit feedback and report incidents or application issues through a built-in form.
- **SR9.2:** The system shall provide customer support with defined response times for user messages.

UR10.0 Map, GPS, and Navigation Services

- **SR10.1:** The system shall detect and use user location through GPS for trip requests and tracking.
- **SR10.2:** The system shall integrate map services to display routes and real-time vehicle movement.
- **SR10.3:** Estimated arrival times shall be calculated based on real-time traffic and GPS data.

Non-Functional Requirements

NFR1.0 Performance

- **NFR1.1:** The app shall load main pages within 3 seconds.
- **NFR1.2:** Real-time location updates shall refresh every 5 seconds.

NFR2.0 Security

- **NFR2.1:** Sensitive data shall be protected using Firebase Authentication, Firebase security rules, role-based access control, OTP verification, and encrypted communication provided by Firebase services.
- **NFR2.2:** The system shall verify user phone numbers using Firebase OTP verification.
- **NFR2.3:** User and trip data shall be protected using role-based access control and secure database rules.

NFR3.0 Reliability

- **NFR3.1:** The system should remain available during normal use when Firebase services and internet connectivity are available.
- **NFR3.2:** Data shall auto-sync when the device reconnects to the internet.

NFR4.0 Usability

- **NFR4.1:** UI design shall follow accessibility standards (clear fonts, contrast).
- **NFR4.2:** New users shall complete registration within 2 minutes.

NFR5.0 Scalability

- **NFR5.1:** The system shall be designed so that routes, users, drivers, and trips can be extended as the service grows.
- **NFR5.2:** Firebase services shall support synchronization of real-time transportation data for the tested project scale.

NFR6.0 Maintainability

- **NFR6.1:** Code shall follow Flutter best practices for modularity.
- **NFR6.2:** Firebase collections, rules, and main service responsibilities shall be organized clearly to support future maintenance.

3.2 Functional Decomposition

This section presents the functional decomposition of the Navigo-وصلني system using use case analysis. It identifies the main actors interacting with the system and summarizes the core use cases that represent the primary system functionalities. The detailed use case specifications are provided in the Appendix.

3.2.1 Actors

The following table presents the primary and secondary actors involved in the Navigo-وصلني system, along with a brief description of their roles.

Primary Actors	
Actor	Description
Passenger	A registered user who can search routes, view nearby vehicles, book seats, track drivers in real time, schedule trips, make payments, and provide feedback.
Driver	A registered driver who can log in, update availability status, accept trip requests, share real-time GPS location, manage assigned trips, and provide feedback.
Route Manager	A staff member responsible for managing buses and microbuses, monitoring drivers, viewing schedules, and supervising daily transportation operations.
Admin	The system administrator who manages all user accounts (passengers, drivers, and route managers), verifies documents, monitors system performance, reviews feedback, resolves issues, and maintains system integrity.

Table 3.1: Primary Actors and Their Roles in the Navigo – وصلني System

Secondary Actors	
Actor	Description
GPS Service	Provides real-time location data for passengers and drivers, enabling live tracking and estimated arrival time (ETA) calculation.
Notification Service	An external service responsible for sending trip alerts, driver arrival notifications, and trip status updates.
SMS Verification Service	Sends One-Time Password (OTP) codes to verify user phone numbers during passenger or driver registration.
Map Service (Google Maps API)	Provides map rendering, route calculation, distance estimation, traffic conditions, and nearby vehicle visualization.

Table 3.2: Secondary Actors and External Services Used in the Navigo – وصالتي System

3.2.2 Use Cases

This subsection presents a summary of the main use cases implemented in the Navigo – وصالتي system. These use cases describe the core interactions between users and the system and are derived directly from the defined user requirements. Full use case specifications, including detailed flows and conditions, are provided in the Appendix.

The system supports the following use cases:

- **UC1: Passenger and Driver Registration**
Allows passengers and drivers to register in the system using role-specific information, verify phone numbers via OTP, and manage profile details. Driver accounts require administrative approval before activation.
- **UC2: Route Browsing and Filtering**
Enables passengers to browse available transportation routes and filter them based on arrival time, route line, destination, price, and vehicle type.
- **UC3: Trip Request**
Allows passengers to request trips using queue-based or direct map-based modes by selecting route, vehicle type, pickup location, and travel time.
- **UC4: Real-Time Tracking**
Enables passengers to track assigned vehicles in real time and view continuously updated estimated arrival times (ETA) during active trips.
- **UC5: Trip Confirmation and History**
Manages trip confirmations and maintains a complete history of completed, canceled, and scheduled trips accessible to passengers, drivers, and administrators.
- **UC6: Notifications and Live Updates**
Provides real-time notifications to passengers and drivers regarding trip status, driver arrival, delays, ETA updates, and payment confirmation, with SMS fallback when required.

- **UC7: Cash Payment Handling**

Allows passengers to complete trip payments using cash, with payment status updates and driver confirmation.

- **UC8: Route and Vehicle Management**

Allows route managers and administrators to manage routes, vehicles, schedules, pricing, and driver assignments through the administrative dashboard.

- **UC9: Safety, Support, and Reporting**

Enables passengers and drivers to report safety issues, misconduct, vehicle problems, and application issues, which are routed to the appropriate responsible role.

- **UC10: Map, GPS, and Navigation Services**

Provides map visualization, GPS-based location tracking, route display, traffic information, and accurate ETA calculation for active trips.

3.2.3 Use Case Specifications

This section includes selected Use Case specifications. The full specifications are available in Appendix A.

- **UC1: Passenger and Driver Registration**

Item	Description
Use Case Name	Passenger and Driver Registration
Actors	Passenger: Creates an account to start using Navigo services. Driver: Creates an account and submits required documents for verification. Admin: Reviews and approves or rejects driver registration requests.
Description	This use case allows new passengers and drivers to register within the Navigo application. Passengers provide basic information such as name and phone number, while drivers submit additional details including vehicle type, plate number, driver license, and assigned route. The system verifies phone numbers using OTP codes, securely stores user profiles, and allows profile updates. Driver accounts require admin approval before activation.
Preconditions	User has the Navigo application installed. User has a valid phone number for OTP verification. Driver has valid documentation (license and vehicle information).
Main Flow	1. User opens the app and selects “Sign Up”. 2. System prompts the user to enter required information based on selected role. 3. User requests OTP verification. 4. System sends OTP to the provided phone number. 5. User enters the OTP. 6. System verifies the OTP successfully. 7. System creates the account: Passenger account is activated immediately. Driver account is created with status “Pending Approval”.
Alternative Flow	Admin reviews pending driver registration profiles. If approved, the driver account is activated. If rejected, the system notifies the driver to resubmit required information. Registered users may edit profile information at any time, and the system validates and saves updates securely.
Exception Flow	Invalid OTP entered: System displays “Incorrect verification code”. Required data missing: System displays “Please fill all required fields”.

Item	Description
Trigger	User selects “Sign Up as Passenger” or “Sign Up as Driver”.
Data	Name, phone number, OTP code, vehicle type, plate number, driver license, assigned route, profile image, admin approval status.
Postconditions	Successful registration: User profile is created and stored (driver remains inactive until approved). Unsuccessful registration: Registration is cancelled or incomplete.
Comments	OTP verification ensures secure identity confirmation. Driver approval workflow enhances safety and platform reliability. Profile data must be stored securely and encrypted.

Table 3.3: Use Case Specification: Passenger and Driver Registration

• **UC2: Route Browsing and Filtering**

Item	Description
Use Case Name	Route Browsing and Filtering
Actors	Passenger: Browses, searches, and filters transportation routes.
Description	This use case allows passengers to browse available transportation routes including buses and microbuses. Passengers can view details such as start and end points, schedule, vehicle type, pricing, seat availability, and ETA. Users may filter routes by arrival time (“Now” or “Scheduled”), route line, price, and destination.
Preconditions	Passenger must be logged into the Navigo system. Route information must exist and be maintained by admin.
Main Flow	1. The user opens the Navigo app and selects “Browse Routes”. 2. The system displays a list of available routes with basic information (vehicle type, start/end points, price, seats, and ETA). 3. The user selects a route to view detailed information.
Alternative Flow	1. The user enters a destination or applies filters (vehicle type, price range, route category). 2. The system updates the displayed list to show only matching routes. 3. The user selects one of the filtered routes to view details.
Exception Flow	No routes match the user’s search or filter criteria; the system displays: “No routes available for the selected criteria.”
Trigger	Passenger selects “Browse Routes” or performs a search/filter action.
Data	Route ID, route name, start point, end point, schedule, price, vehicle type, seat availability, ETA, filters, search query.
Postconditions	Successful browsing: Passenger views route/vehicle details and may proceed to request a trip. No results: Passenger is informed that no suitable routes are available.
Comments	Real-time seat availability and ETA should be updated frequently. The map should highlight the selected route path when viewing details. Admins and route managers must ensure data accuracy to avoid user confusion.

Table 3.4: Use Case Specification: Route Browsing and Filtering

• **UC3: Trip Request**

Item	Description
Use Case Name	Trip Request
Actors	Passenger: Requests trips and tracks assigned vehicles. Driver: Receives direct trip requests (microbus) and accepts or declines them.
Description	This use case allows passengers to request a trip using two modes: (1) queue-based requests for microbus or bus routes managed by a route manager, and (2) direct map-based requests sent to active vehicles such as microbuses. The system displays estimated fare, ETA, vehicle type, and seat availability. After assignment, the passenger can track the driver's live location until pickup. Drivers may accept or decline direct requests, and passengers may cancel or modify their request before the trip begins.
Preconditions	Passenger is logged in. GPS and map services are available. For queue-based mode: a valid microbus/bus route exists and is managed by a route manager. For direct mode: at least one active microbus is available on the map. Driver accounts are active and online.
Main Flow	1. The passenger selects route, vehicle type (bus, microbus, taxi, etc.), and desired travel date and time. 2. The system searches for available trips that exactly match the selected criteria. 3. If exact matches are found, the system assigns the passenger to the appropriate trip (queue-based or direct), notifies the passenger and assigned driver, and enables live tracking and ETA display. 4. If no exact matches are found, the system suggests alternative trips (e.g., different time slots, similar vehicle types, or nearby routes). 5. The passenger selects an alternative trip, and the system proceeds with assignment and notifications.

Item	Description
Alternative Flow	1. The passenger selects route, vehicle type, date, and time. 2. The system finds no exact match and generates alternative trips based on nearby departure times, similar vehicle types, and nearby or related routes that serve the destination. 3. The system displays the suggested alternatives with estimated fare, ETA, and seat availability. 4. The passenger selects a preferred option, and the system assigns the trip accordingly. 5. If the passenger rejects all alternatives or cancels, the system ends the request and displays: “No suitable trips available at this time. Please try again later or adjust your search criteria.”
Exception Flow	No available vehicles are found (microbuses, buses, or taxis); the system displays: “No available vehicles now. Please try again later.”
Trigger	Passenger selects “Request Trip” or “Request Seat” on a route.
Data	Route ID, passenger ID, driver ID, vehicle ID, pickup location, destination, ETA, estimated fare, seat availability, queue position, trip status, cancellation reason.
Postconditions	Accepted trip: A trip is created and linked to the assigned vehicle and passenger, and the passenger can track the driver’s live location. Declined or cancelled trip: No active trip exists, and both parties receive appropriate notifications.
Comments	Queue-based assignment is best for fixed microbus/bus routes. Direct map-based requests support flexibility for microbuses/taxis already on the road. Real-time tracking should update frequently to keep ETAs accurate.

Table 3.5: Use Case Specification: Trip Request

3.2.4 Use Cases Diagram

The Use Case Diagram shows the main user interactions with the Navigo-وصلي system, highlighting key actors and use cases to define system boundaries and core functionality.

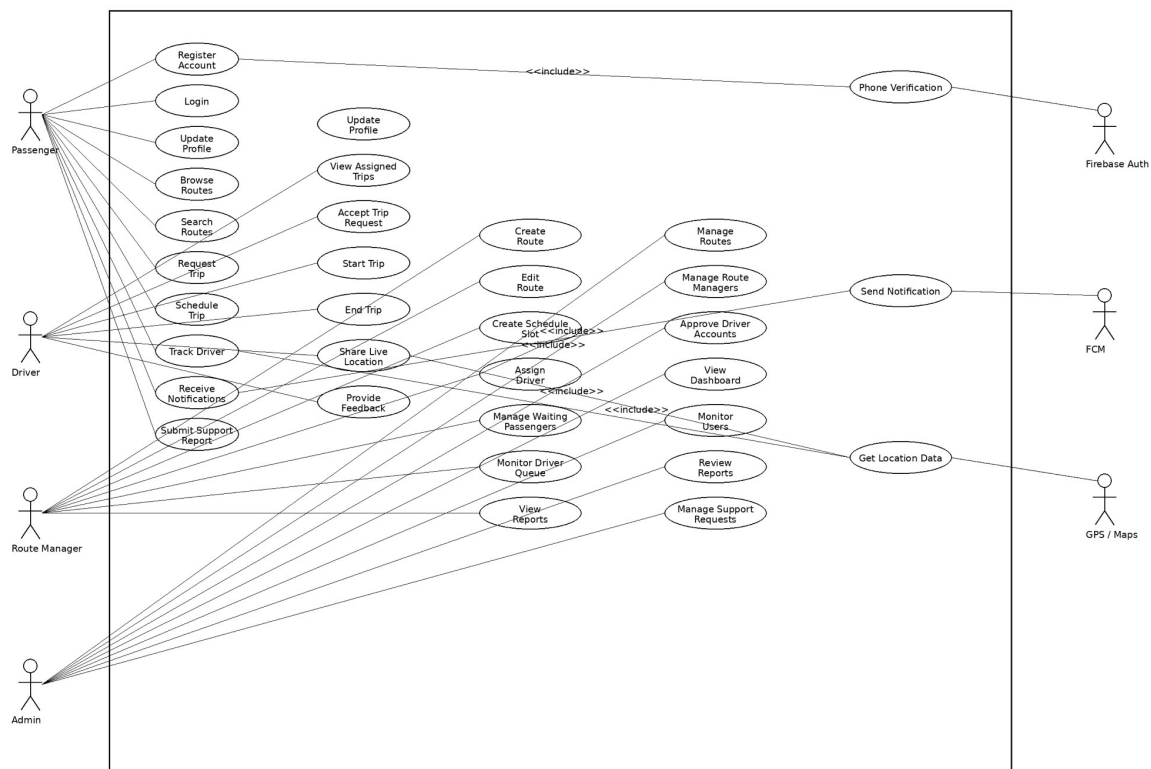


Figure 3.1: Use Case Diagram for the Navigo – وصلي System

3.3 System Models

This section presents the main system models used to describe the structural and behavioral aspects of the Navigo-وصلي transportation platform. These models provide a clear understanding of how system components are organized, how actors interact with the system, and how system states evolve during operation. The models include class, sequence, activity, and state chart diagrams.

3.3.1 Class Diagram

The class diagram shows the main system classes, their attributes/operations, and relationships in Navigo-وصلي (e.g., Passenger, Driver, Trip, Route, Vehicle, Payment, Notification).

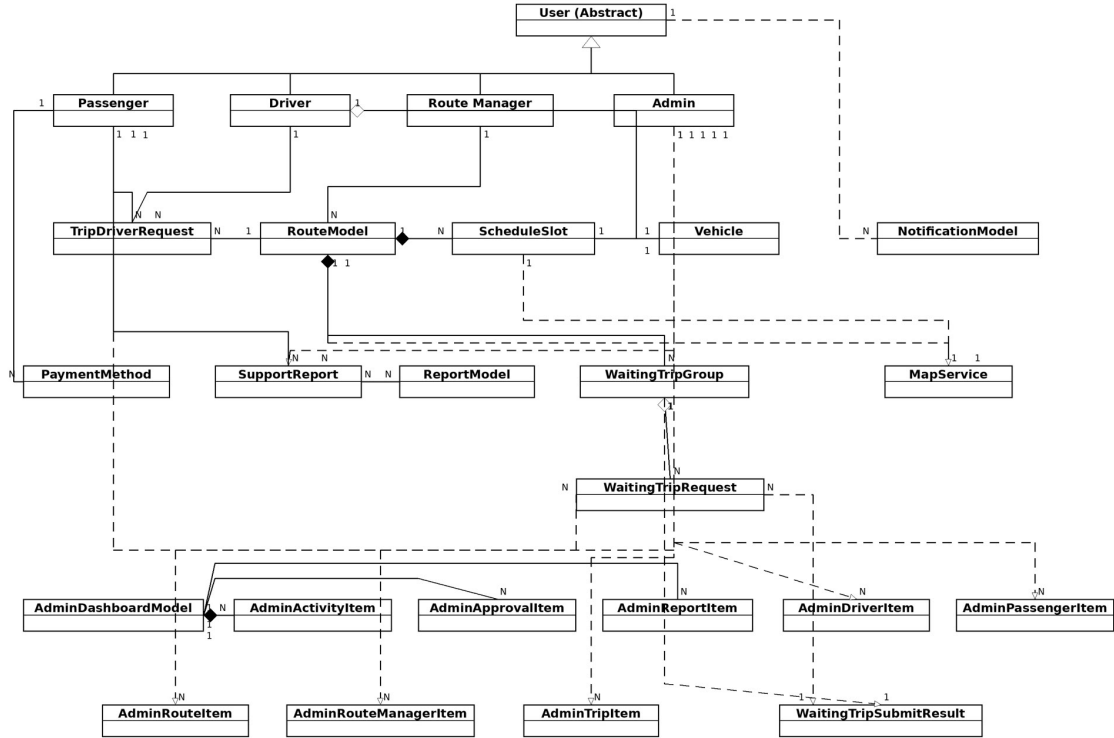


Figure 3.2: General Class Diagram for the Navigo – وصلي System

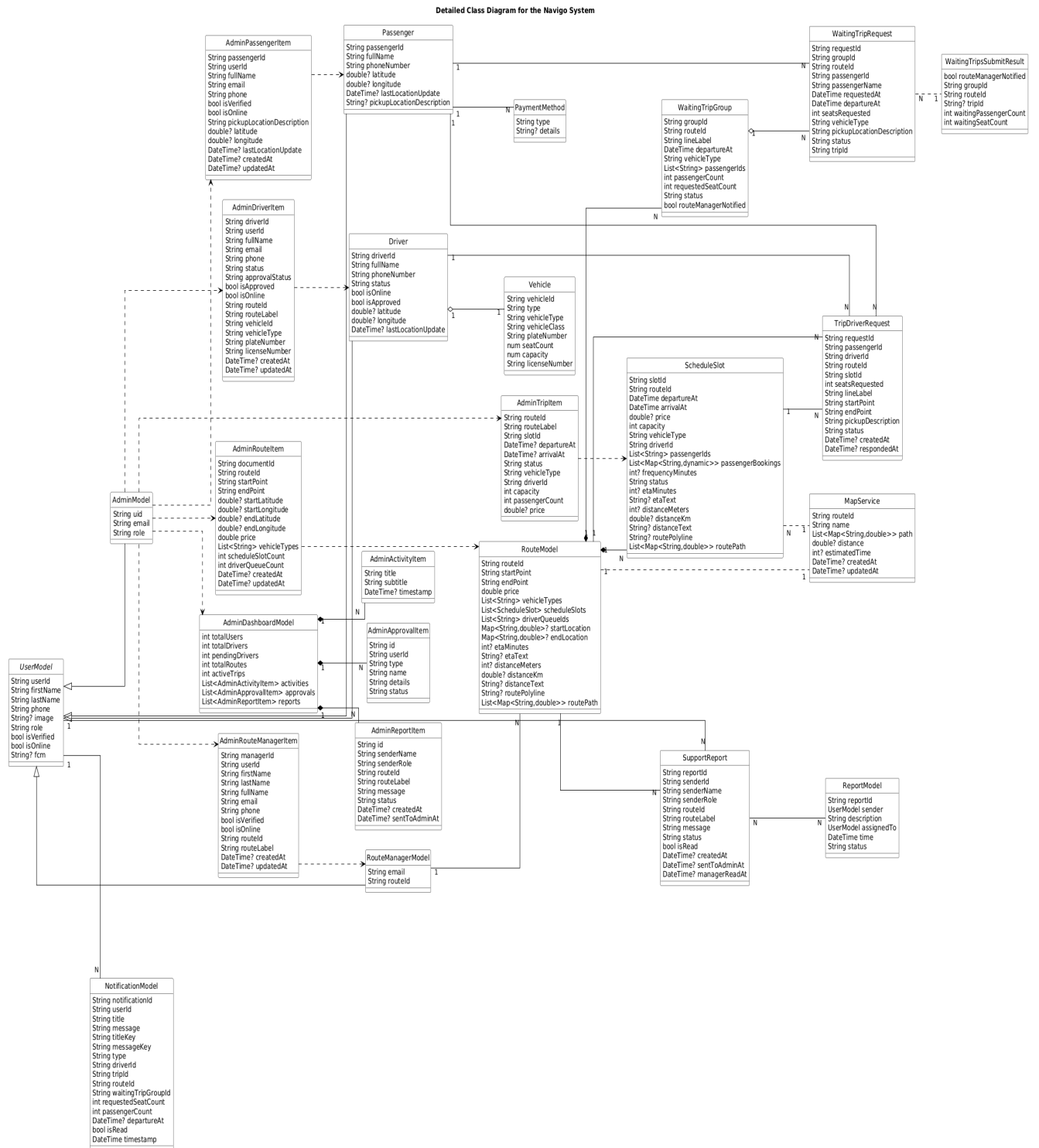


Figure 3.3: Detailed Class Diagram for the Navigo – وصلي System

3.3.2 Sequence Diagram

The sequence diagrams show the time-ordered interactions between users, the app, and backend services for key scenarios (registration, route browsing/filtering, and trip requests), including decision and alternative flows.

Passenger and Driver Registration (UR1.0)

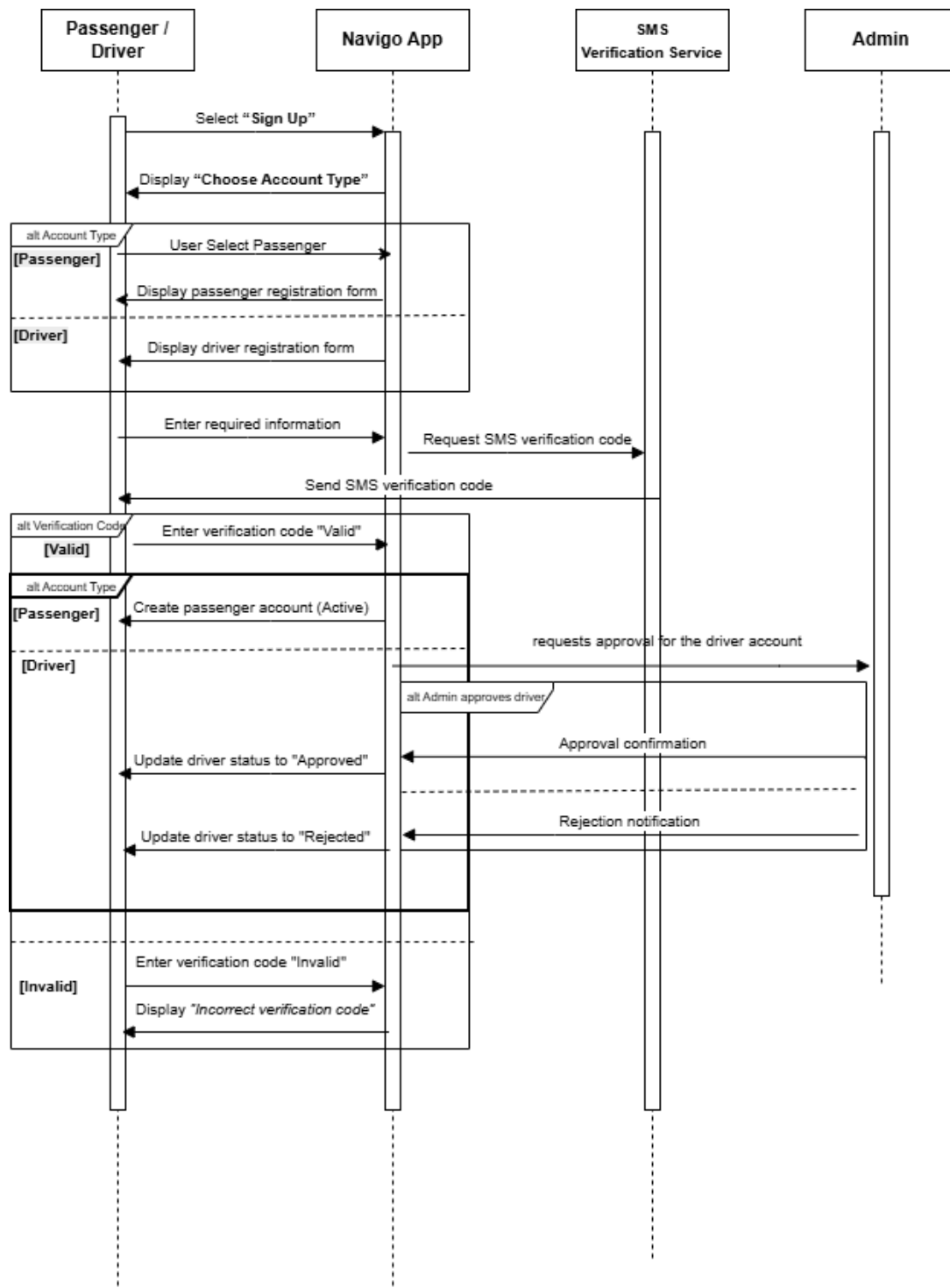


Figure 3.4: Sequence Diagram for Passenger and Driver Registration (UR1.0)

Route Browsing and Filtering (UR2.0)

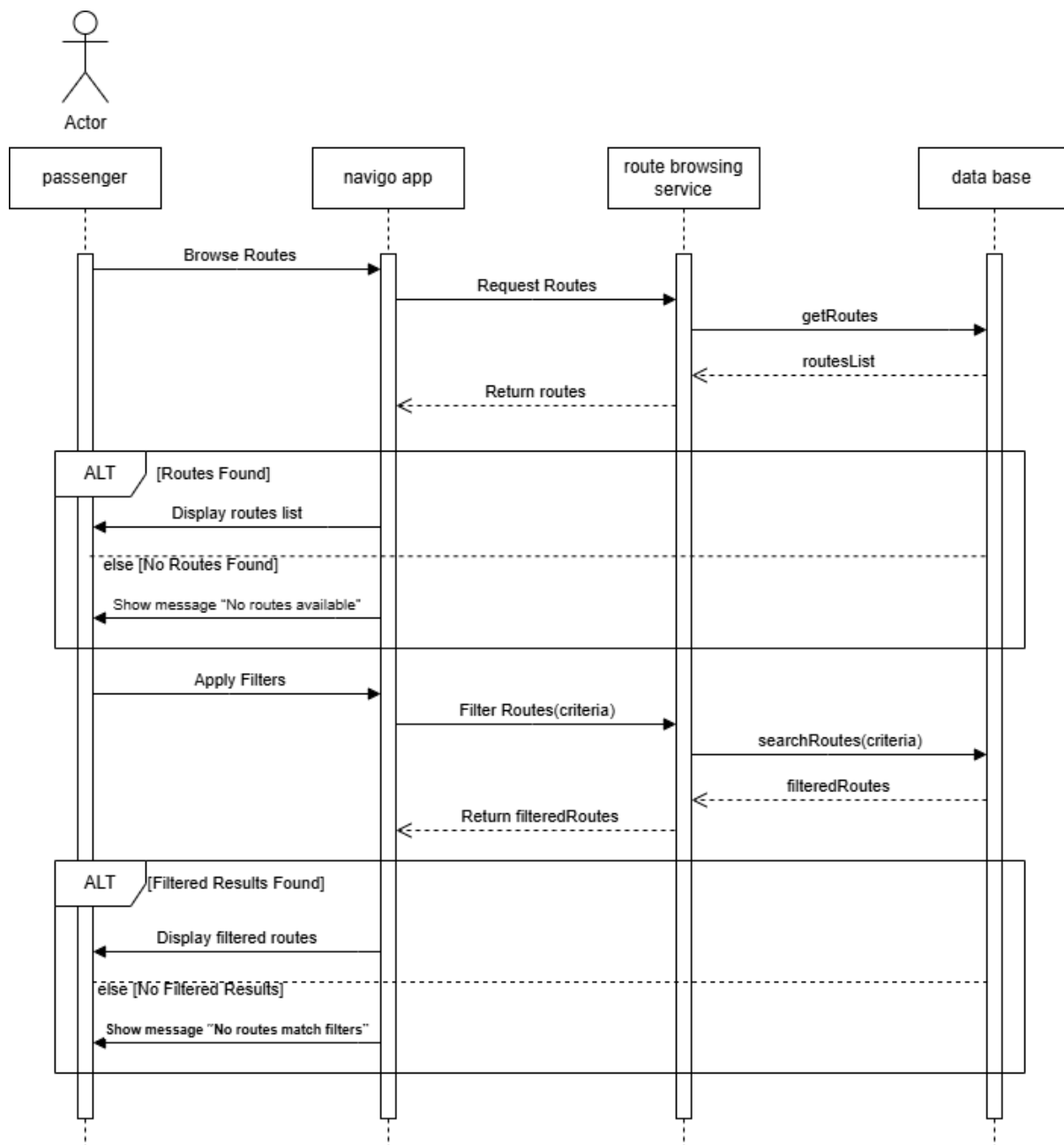


Figure 3.5: Sequence Diagram for Route Browsing and Filtering (UR2.0)

Trip Request Processing (UR3.0)

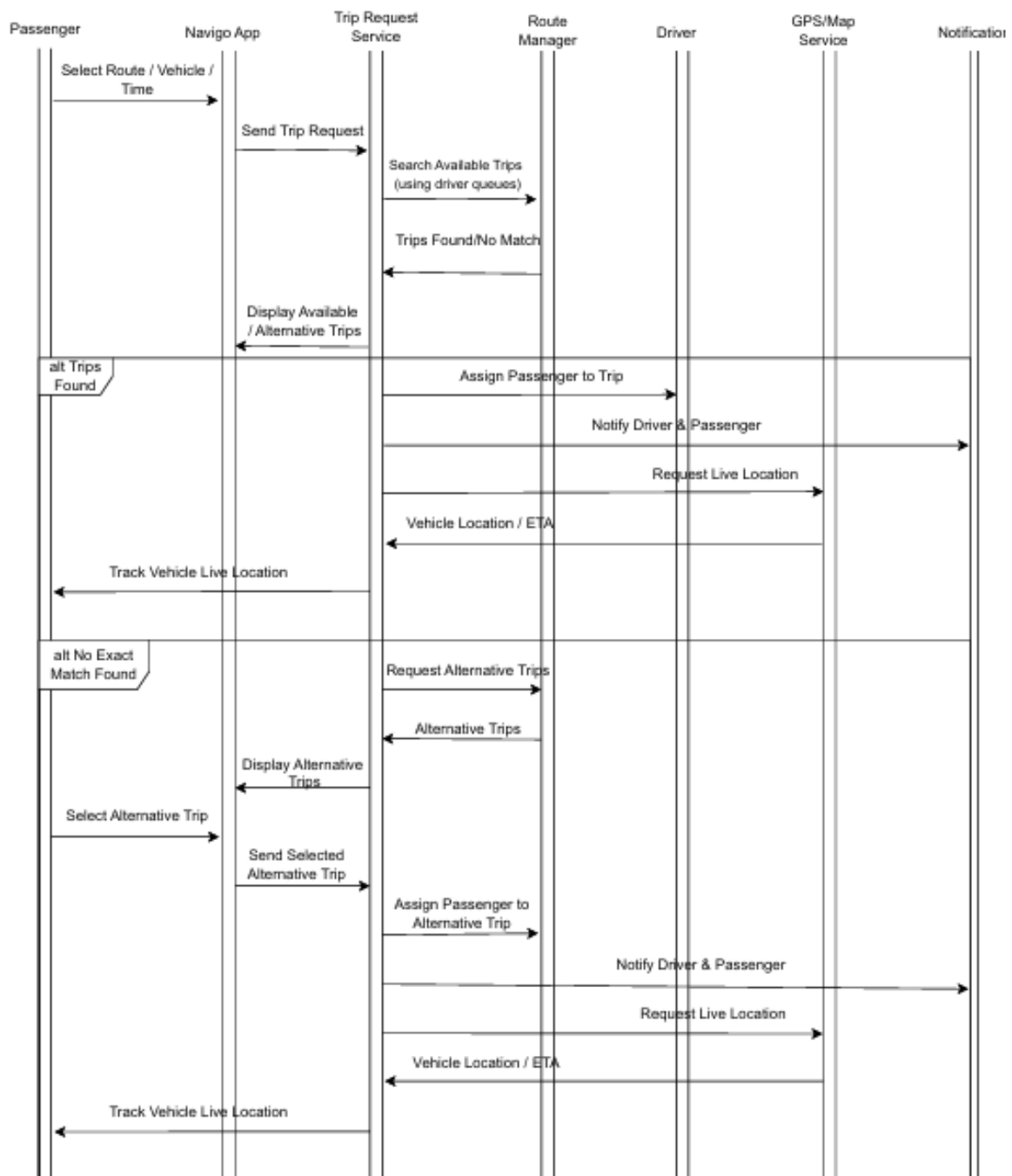


Figure 3.6: Sequence Diagram for Trip Request Processing (UR3.0)

3.3.3 Activity Diagram

A consolidated activity diagram presents the end-to-end workflow of Navigo-وصلي, from registration and authentication to route selection, trip request/tracking, and cash payment handling, including key decision points (roles, approvals, availability, OTP, and payment status).

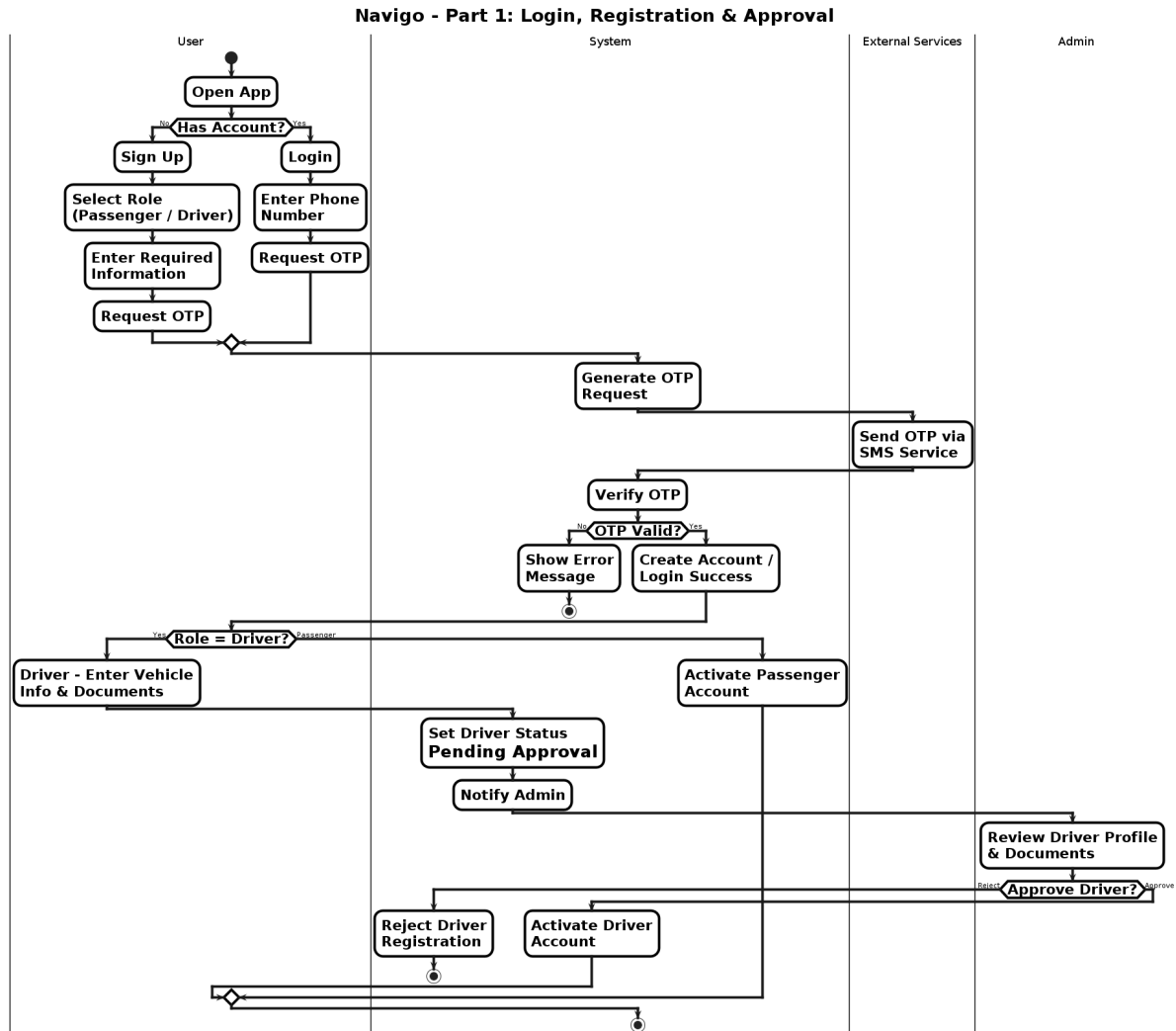


Figure 3.7: Part 1 of the Activity Diagram for the Navigo-وصلي System

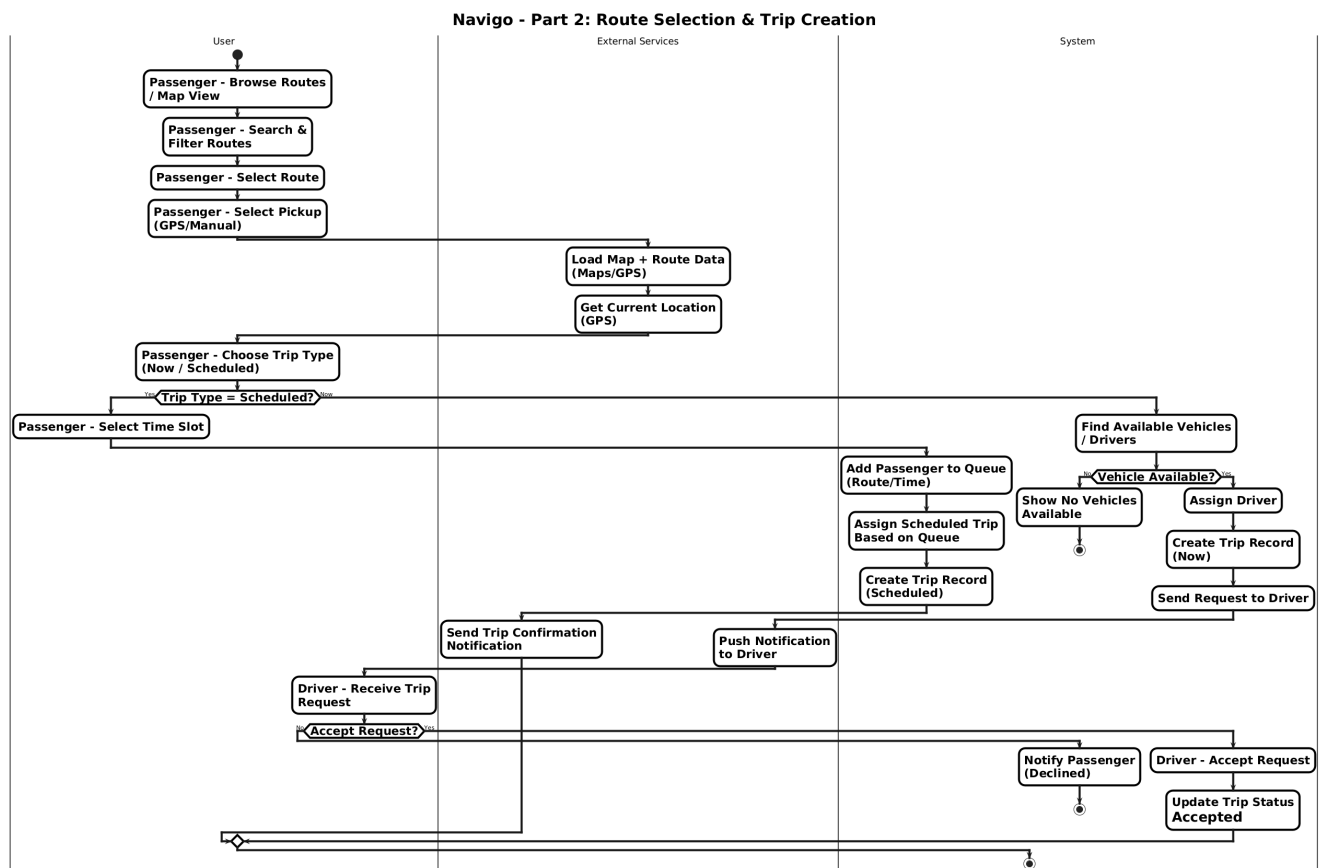


Figure 3.8: Part 2 of the Activity Diagram for the Navigo-وصلني System

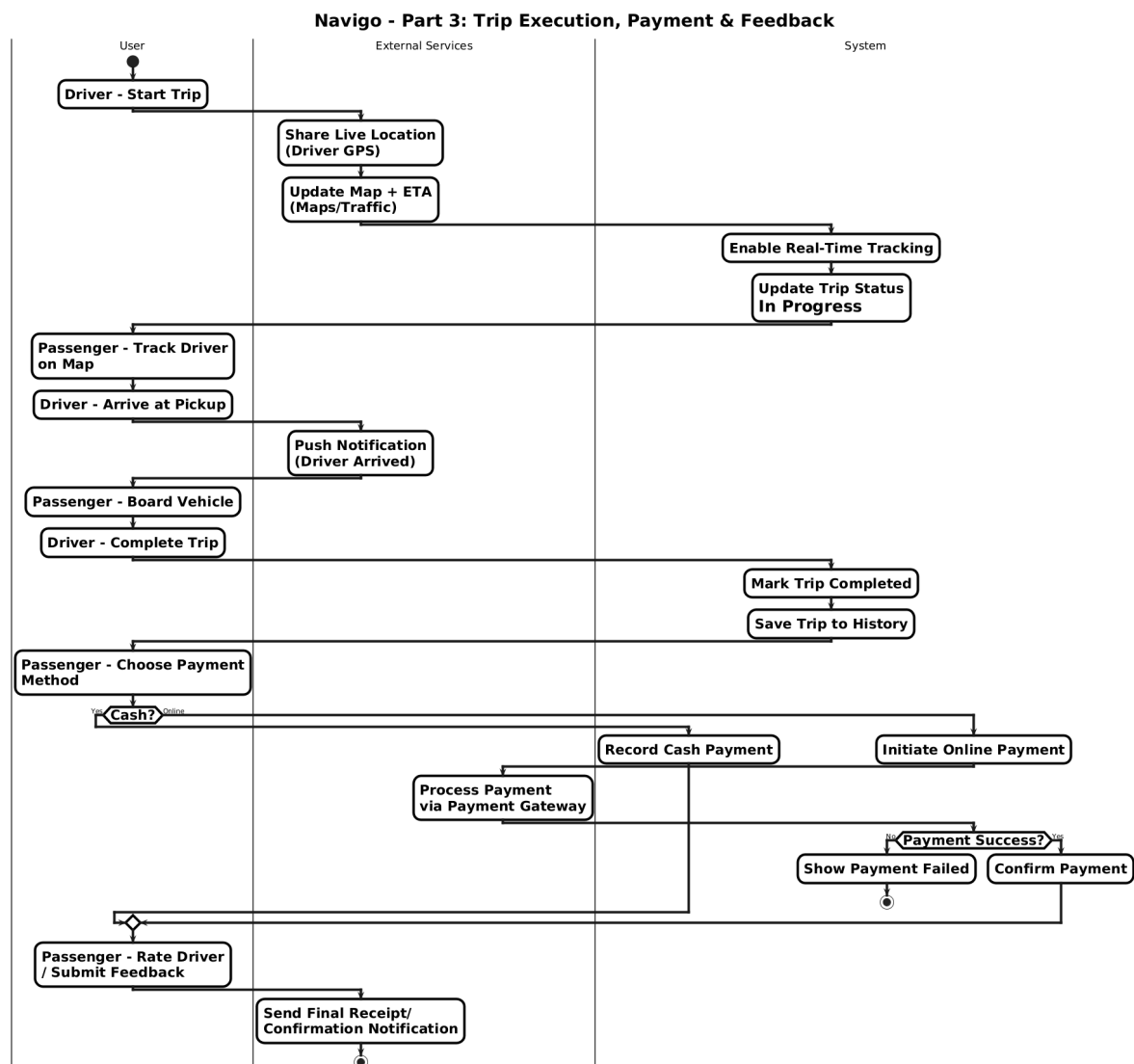


Figure 3.9: Part 3 of the Activity Diagram for the Navigo-وصلي System

3.3.4 State Chart Diagram

The state chart diagram models the trip lifecycle in Navigo-وصّلي, showing key states and transitions from creation to completion or cancellation, including exceptions such as rejection and payment failure.

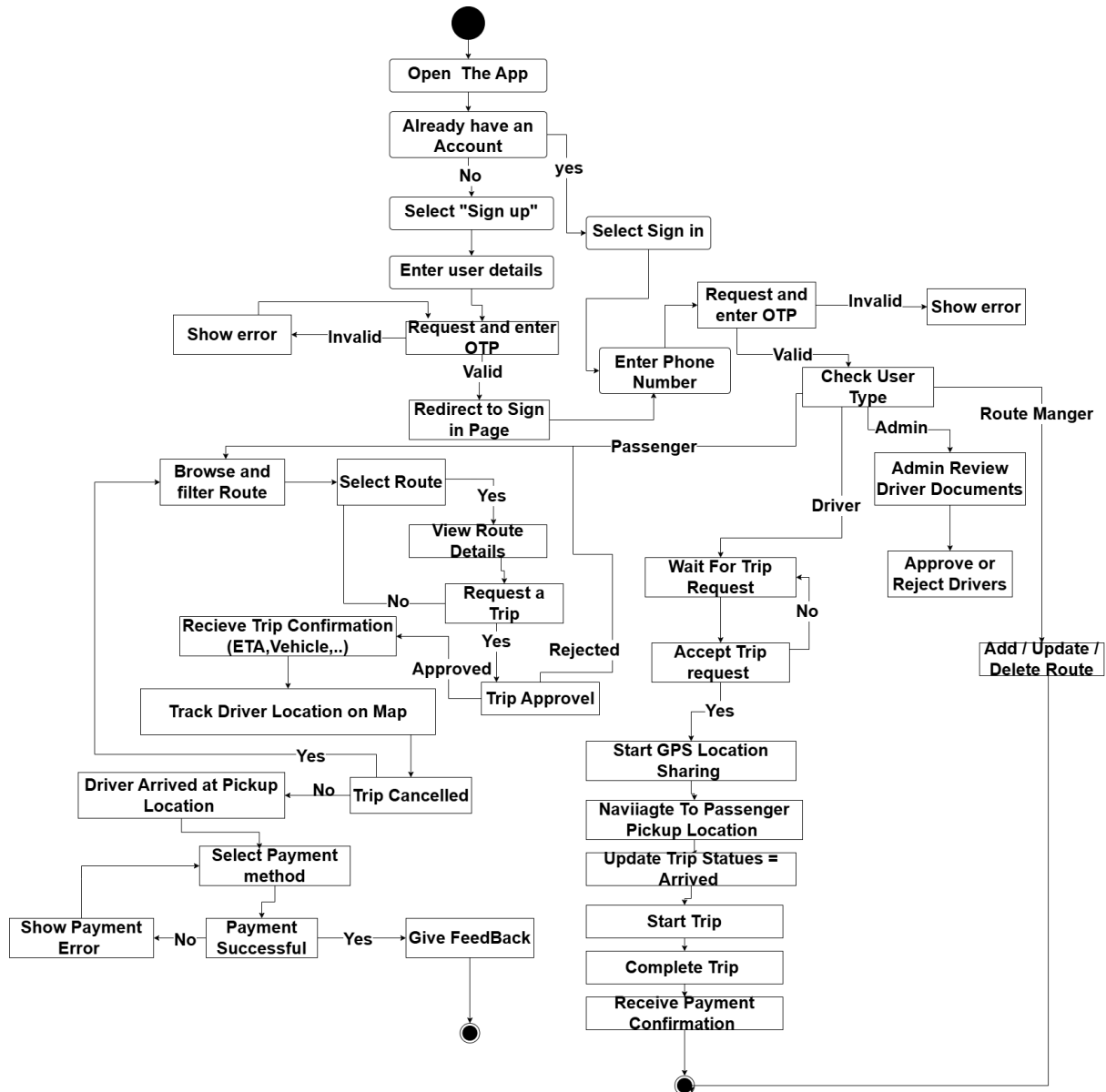


Figure 3.10: State Chart Diagram for Trip Lifecycle in the Navigo-وصّلي System

3.4 System Architecture

This section presents the overall architecture of the Navigo-وصلتي system. It describes how the system is organized into logical subsystems, the software architectural style used to implement system functionality, and the deployment of software components across hardware and network infrastructure. The architecture is designed to ensure scalability, reliability, security, and efficient communication between users, services, and external systems.

3.4.1 Sub-System

The Navigo-وصلتي platform is divided into several interacting subsystems, where each subsystem provides a focused set of services. This modular structure improves maintainability, scalability, and separation of concerns. The main subsystems are:

- **User Management Subsystem:** Supports passenger and driver registration, OTP-based authentication, profile management, and role-based access control.
- **Route Management Subsystem:** Manages route definitions and operational data such as route lines, schedules, pricing, seat availability, and route updates. It also supports route creation, editing, and deletion through the admin/route manager panel.
- **Trip Management Subsystem:** Handles trip request creation, trip assignment to available drivers/vehicles, trip confirmation, status updates (started, ended), cancellations, and trip history for both passengers and drivers.
- **Tracking Subsystem:** Provides real-time tracking of drivers/vehicles during active trips, updates driver status (Available, On Trip, Offline), and maintains tracking records for monitoring and analytics.
- **Map and Navigation Subsystem:** Integrates map services to display routes and vehicle movement, supports GPS-based pickup selection, route visualization, and ETA estimation based on traffic and distance.
- **Payment Subsystem:** Manages cash payment status for trips, including estimated fare display, cash-pending status, and driver confirmation after collection.
- **Notification Subsystem:** Delivers real-time updates such as trip acceptance, driver arrival, delays, cancellations, and payment confirmation using push notifications.
- **Administration and Reporting Subsystem:** Allows admins to manage routes, vehicles, and driver accounts, and handle user feedback, safety reports, and issue resolution.
- **Database and Data Management Subsystem:** Stores and manages system data such as users, drivers, vehicles, routes, trips, payments, notifications, and reports. It ensures data integrity, access control, and supports analytics and logging.

3.4.2 Software Architecture

Navigo-وصلني uses a layered architecture that separates UI (passenger/driver/route manager apps and admin panel), application services (routes, trips, auth, tracking/ETA, cash payment status, notifications, safety/support), and infrastructure/data (database and external services such as maps/GPS, OTP, and notifications).

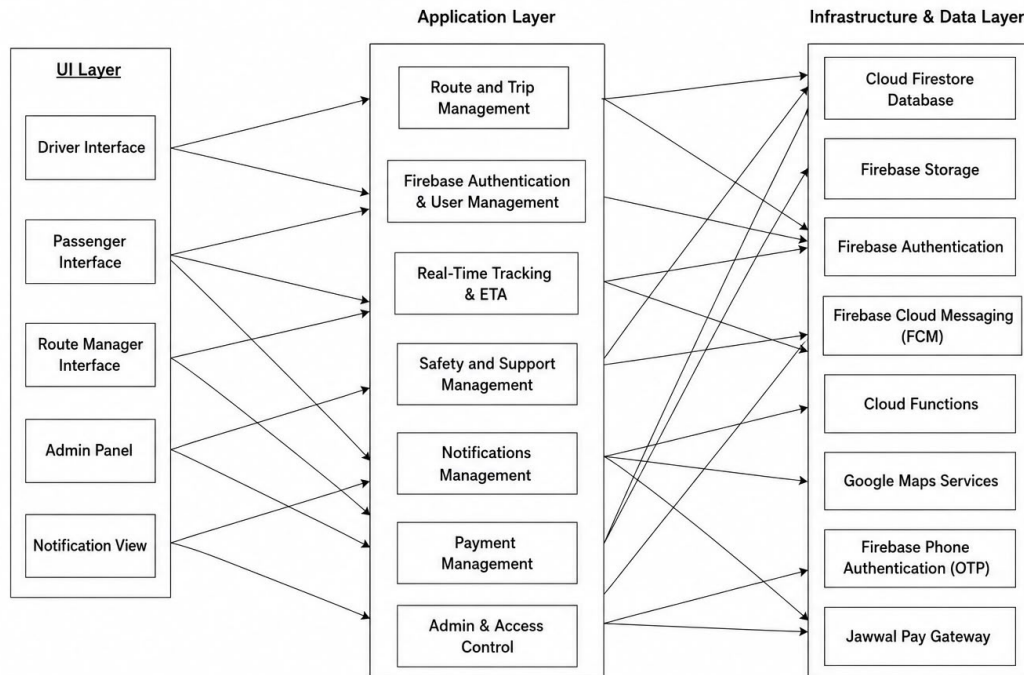


Figure 3.11: Architecture Diagram for the Navigo-وصلني System

3.4.3 Deployment Diagram

The deployment diagram shows Navigo-وصلني components across client devices (Passenger, Driver, route manager, Admin) connected to an application server that provides core services and integrates with external services such as the database, notifications, and maps/GPS. Cash payment is handled operationally by the driver and does not require a digital payment gateway in the implemented system.

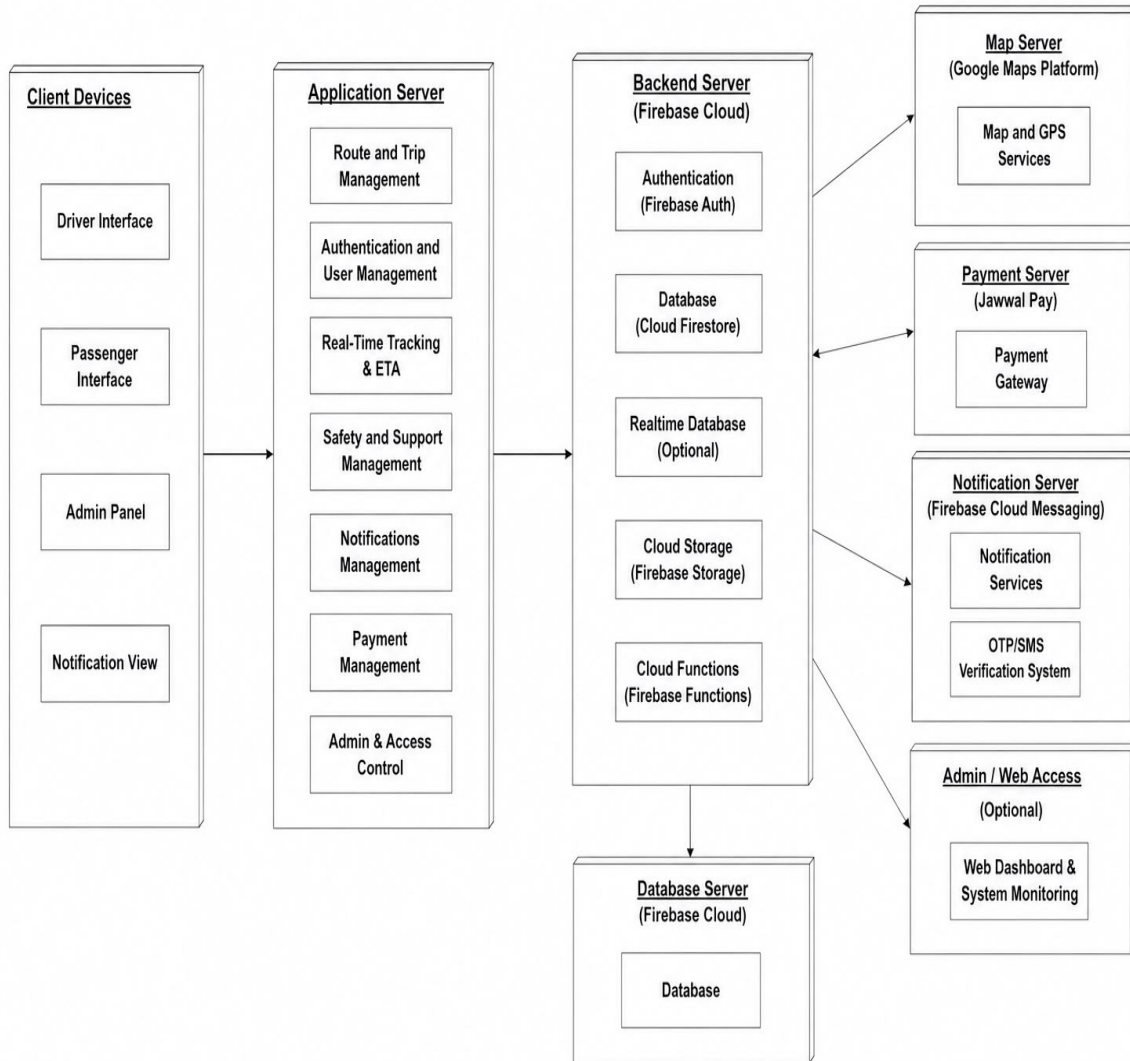


Figure 3.12: Deployment Diagram for the Navigo-وصلي System

3.5 Data Management and ERD

This section outlines the Navigo-وصلي data model, where a centralized database stores users, routes, trips, vehicles, payments, notifications, and reports, with integrity and security enforced via entity relationships and role-based access control. The ERD summarizes these entities and relationships.

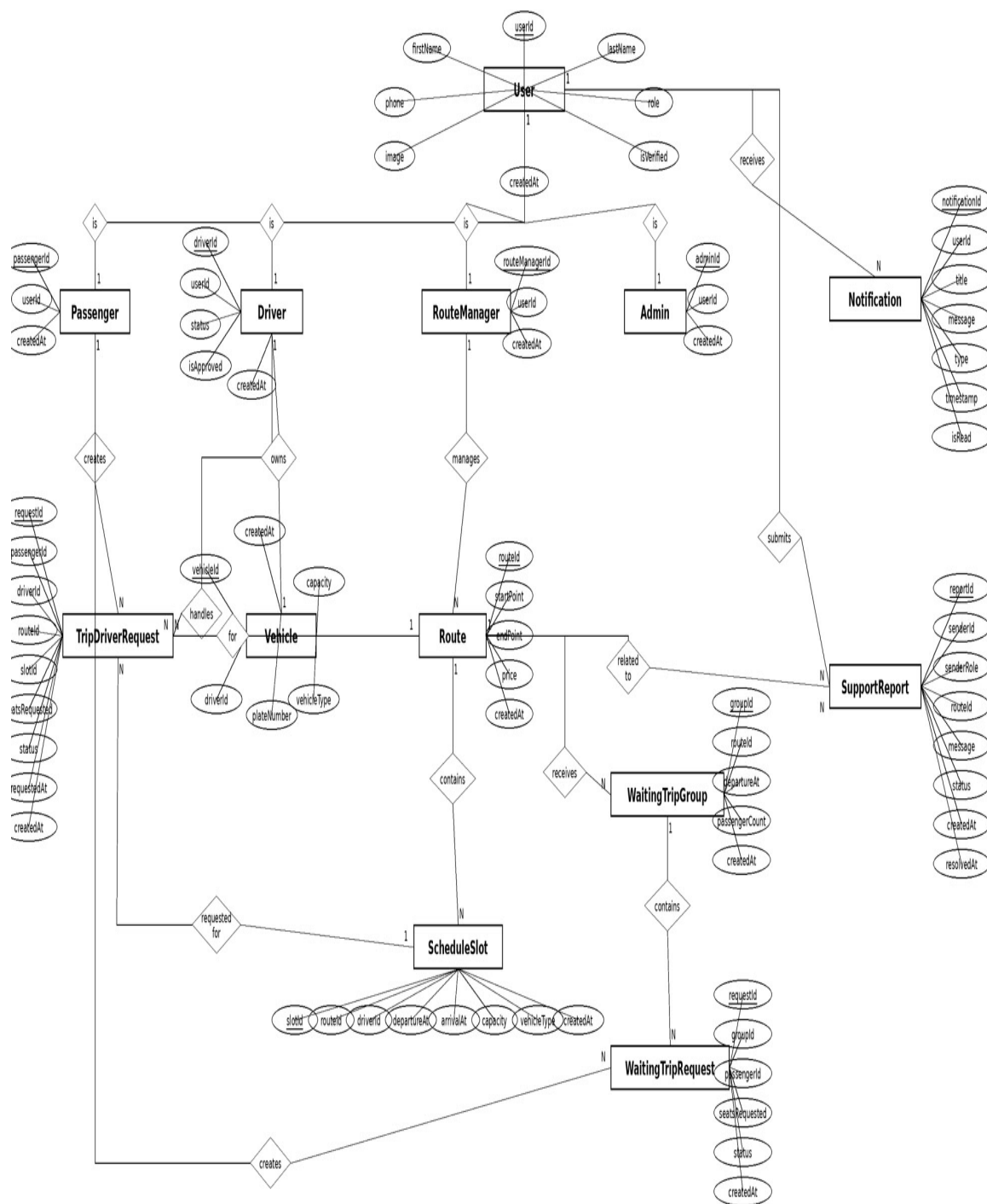


Figure 3.13: Entity-Relationship Diagram (ERD) for the Navigo-وصلي System

3.6 UI Design and Figma Prototype

This section provides a mini prototype of the system interfaces to offer a high-level overview of the user experience. For more screens click [15].

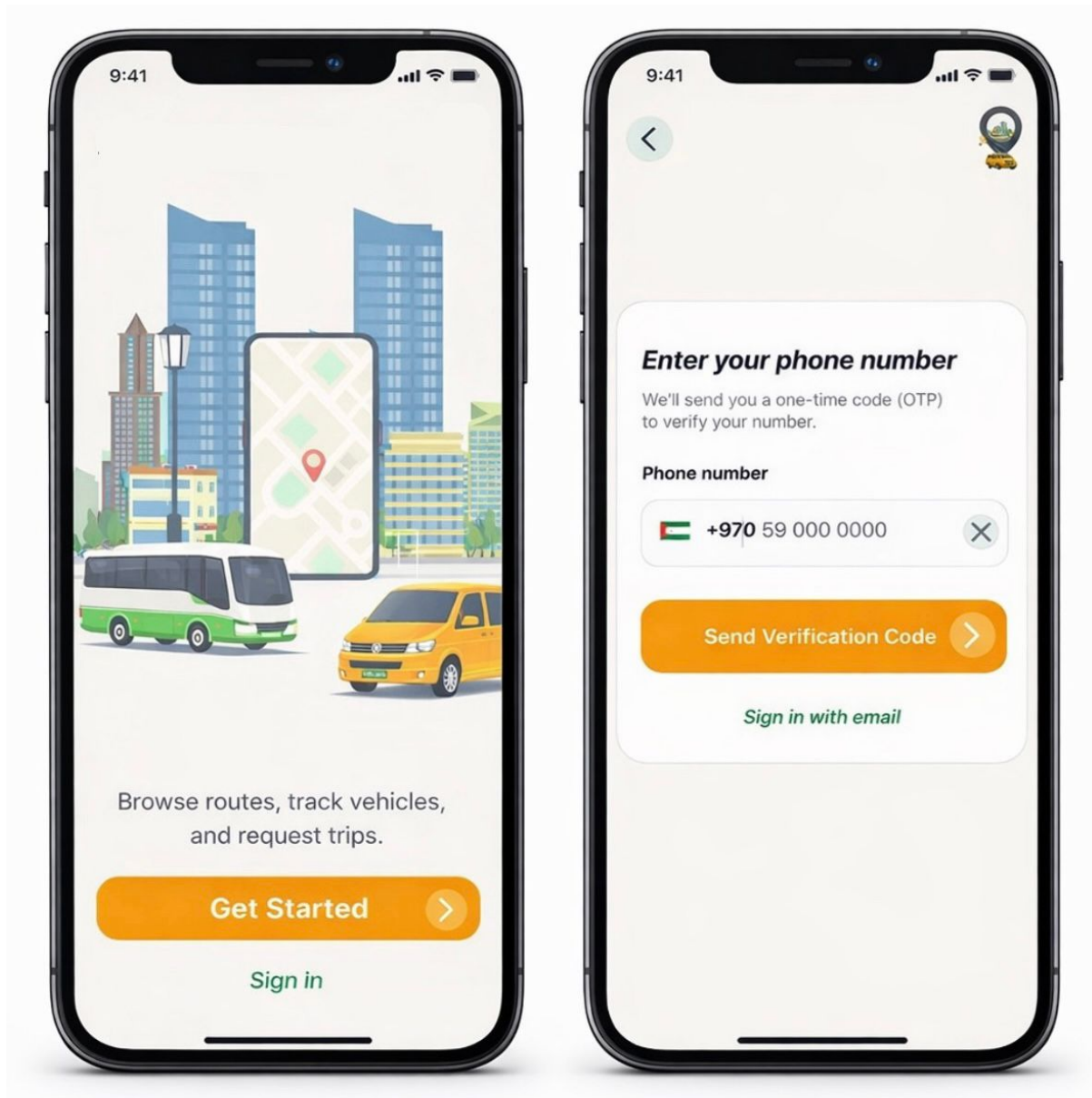


Figure 3.14: Sign in Interface

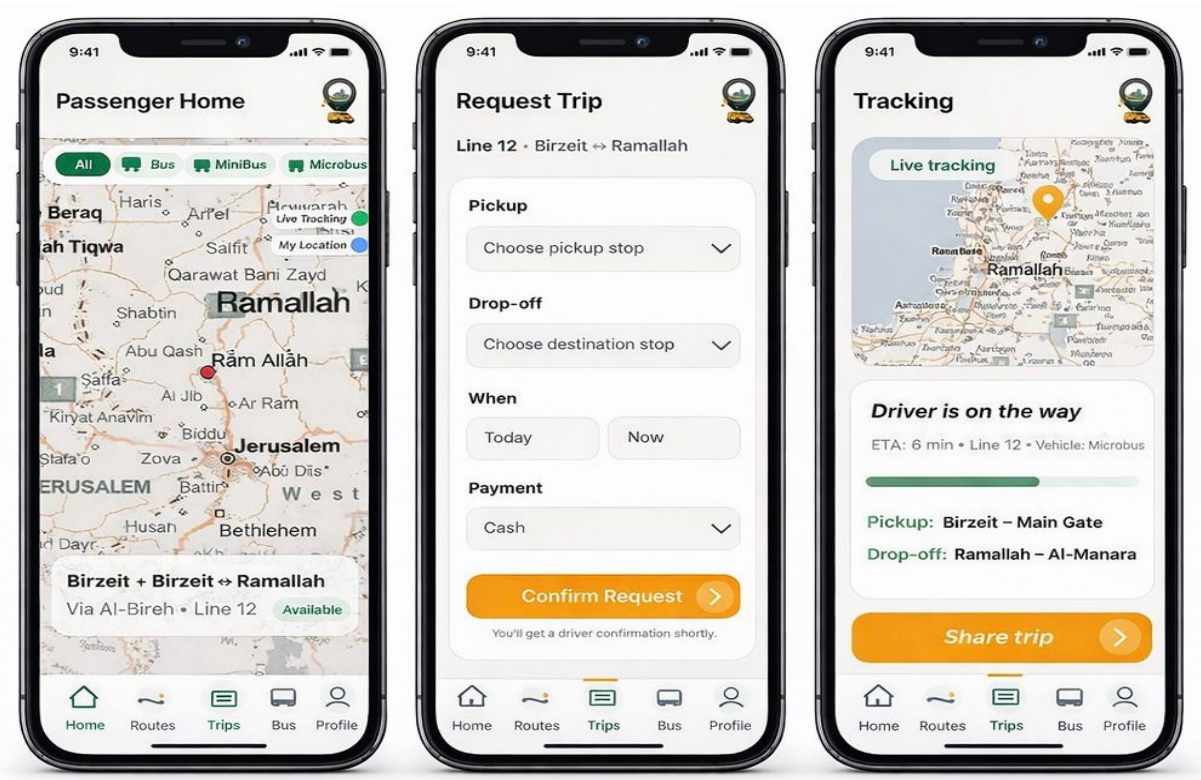


Figure 3.15: Passenger Interface

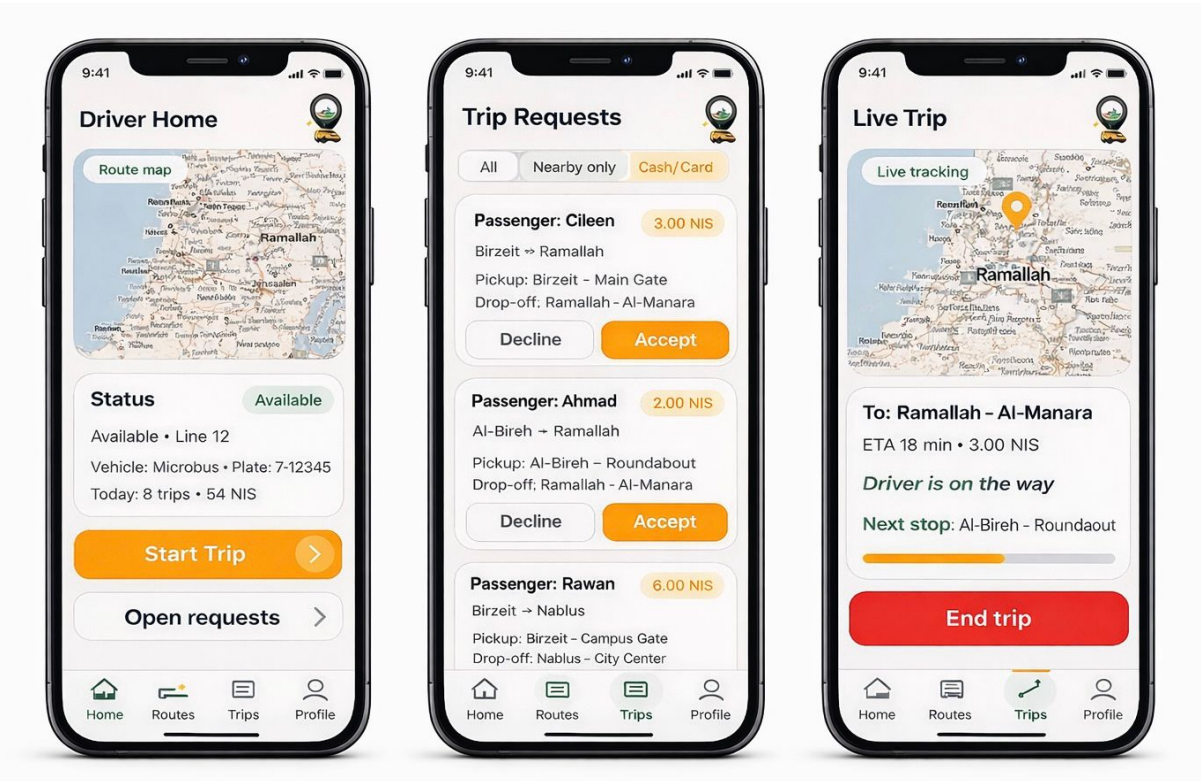


Figure 3.16: Driver Interface

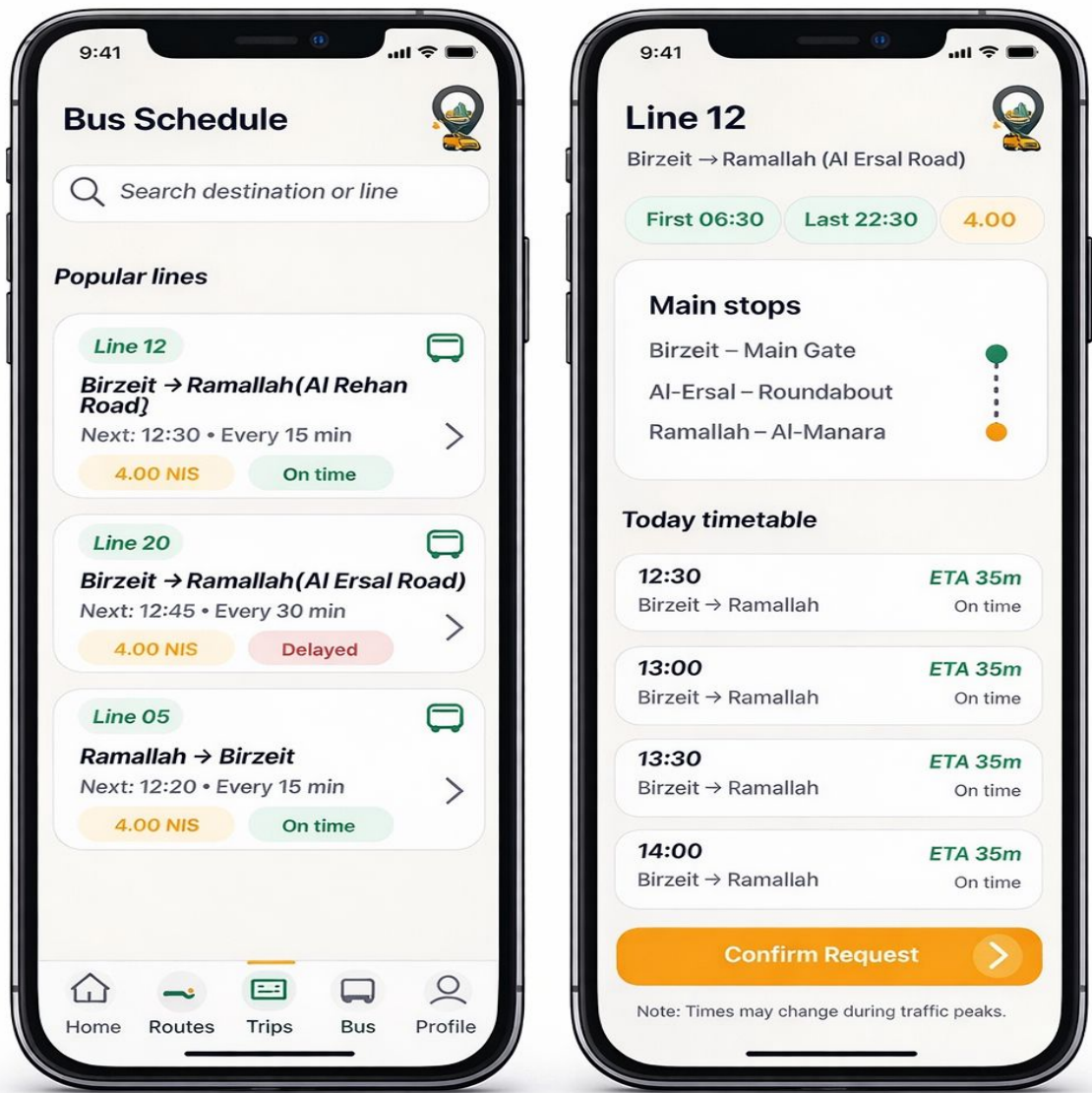


Figure 3.17: Schedule bus Interface

Chapter 4

Implementation

The screenshots presented in this chapter are taken from the implemented Navigo-وصلني application and administrative interfaces.

4.1 User Registration and Authentication

This section presents the implementation of the registration and authentication workflows in the Navigo application. It covers language and role selection, passenger and driver registration, OTP verification, and login interfaces for users, Route Managers, and administrators.

4.1.1 Welcome and Language Selection

The application starts with a welcome screen that allows users to select their preferred language, Arabic or English, before continuing. This improves accessibility and provides a better user experience for different users.

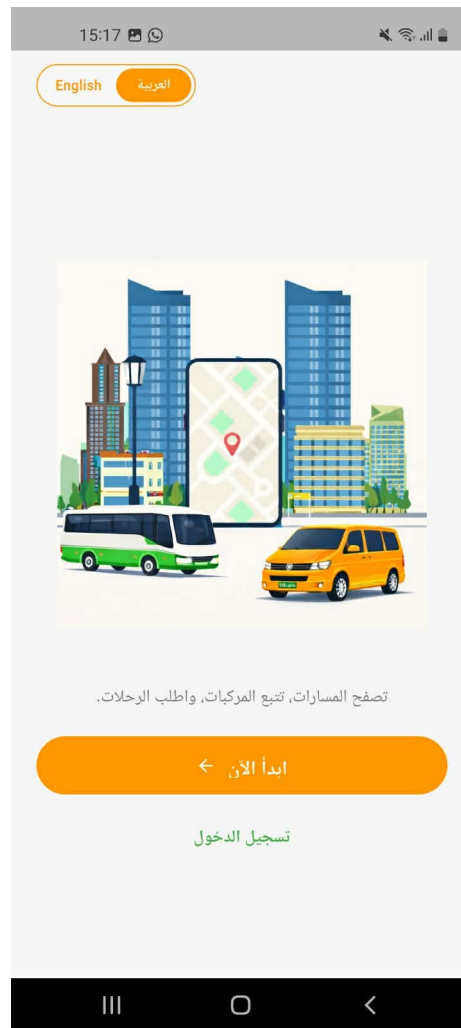


Figure 4.1: Welcome and language-selection screen

4.1.2 User Role Selection

After selecting the language, users choose whether they want to register as a passenger or a driver. Each role is directed to a dedicated registration process based on its requirements.

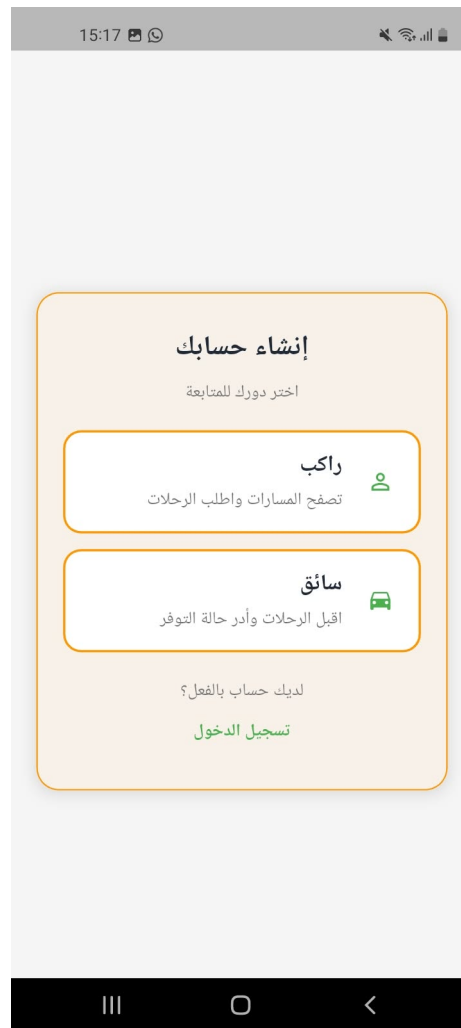
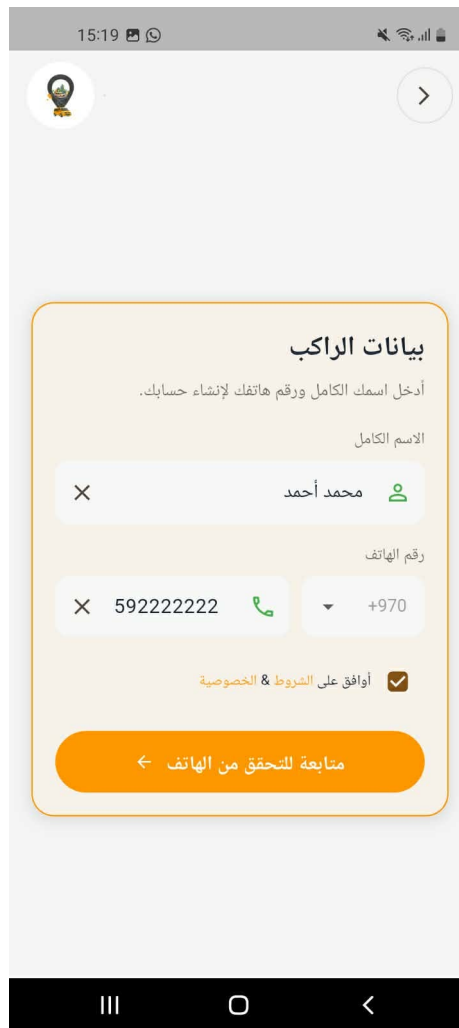


Figure 4.2: Passenger and driver role-selection screen

4.1.3 Passenger Registration

Passengers can create an account by entering their full name and phone number and agreeing to the terms and conditions. The provided information is validated before proceeding to phone verification.



The image shows a mobile application interface for passenger registration. At the top, there is a status bar with the time 15:19 and various icons. Below the status bar, there is a header area with a circular profile picture placeholder on the left and a right arrow icon on the right. The main content area is a light gray rectangle containing a registration form. The form has a title "بيانات الراكب" (Passenger Data) and a subtitle "أدخل اسمك الكامل ورقم هاتفك لإنشاء حسابك." (Enter your full name and phone number to create your account). The form contains two input fields: "الاسم الكامل" (Full Name) with the text "محمد أحمد" (Mohammed Ahmed) and "رقم الهاتف" (Phone Number) with the text "592222222" and a country code dropdown set to "+970". There is a checkbox labeled "أوافق على الشروط & الخصوصية" (I agree to the terms & privacy) which is checked. At the bottom of the form is a large orange button with the text "متابعة للتحقق من الهاتف" (Continue to verify phone). The bottom of the screen shows a black navigation bar with three icons: a hamburger menu, a circle, and a back arrow.

Figure 4.3: Passenger registration screen

4.1.4 OTP Verification

To verify ownership of the phone number, a one-time password (OTP) is sent to the user's device. The user must enter the received code to complete the verification process and activate the account.

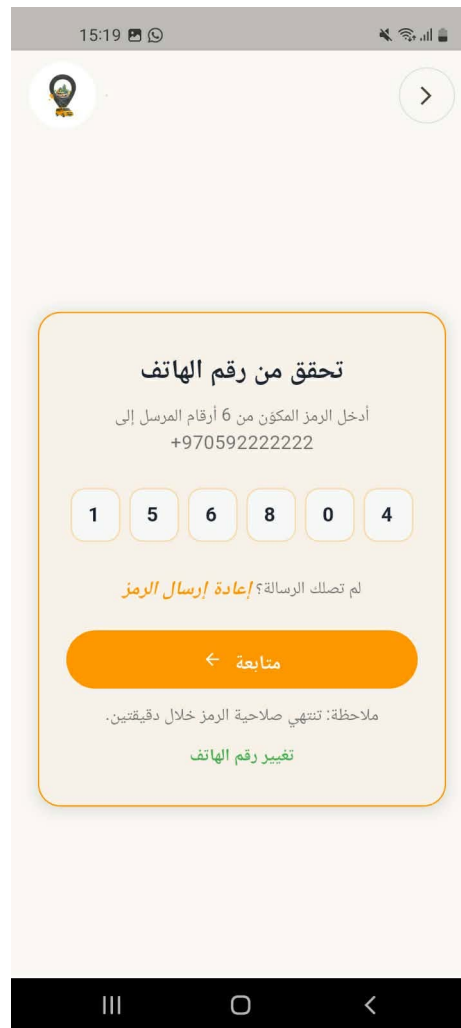


Figure 4.4: OTP verification screen

4.1.5 Driver Registration

Driver registration requires additional information compared with passenger registration. Drivers must provide their full name, phone number, assigned route, vehicle information, and other required details.

15:21

بيانات السائق

الاسم الكامل

× عمر يوسف

رقم الهاتف

× 593333333 +970

المسار (من Firestore)

↓ Ramallah → Birzeit

رقم المركبة (اللوحة)

× 5-1549

رقم رخصة القيادة (اختياري)

× رقم الرخصة

نوع المركبة

↓ Bus - 45 seats

أوافق على الشروط & الخصوصية ✓

Figure 4.5: Driver registration screen

4.1.6 Driver Request Confirmation

Once the driver submits the registration form, the system displays a confirmation message indicating that the request has been received successfully and is awaiting administrative approval.

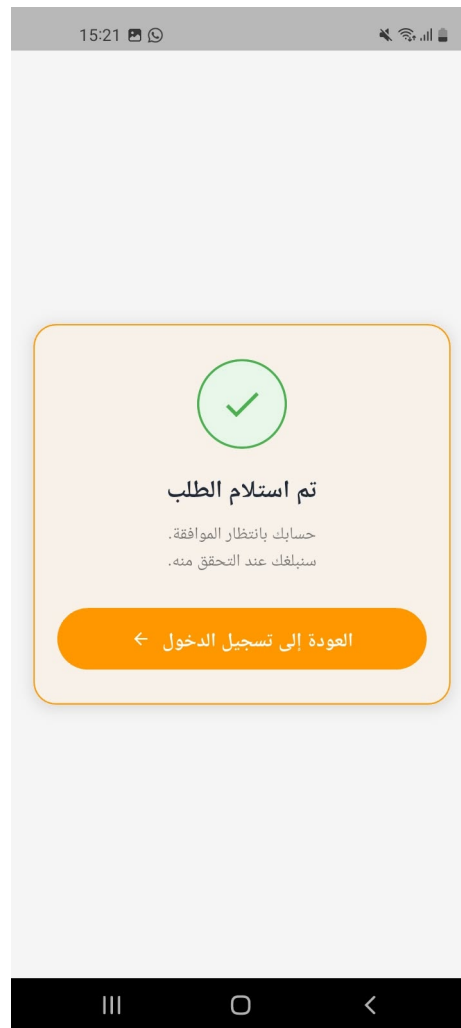


Figure 4.6: Driver registration-request confirmation

4.1.7 Route Manager Login

Route Managers access the system using their assigned email address and password. This login process allows them to manage routes, schedules, and trip operations according to their permissions.

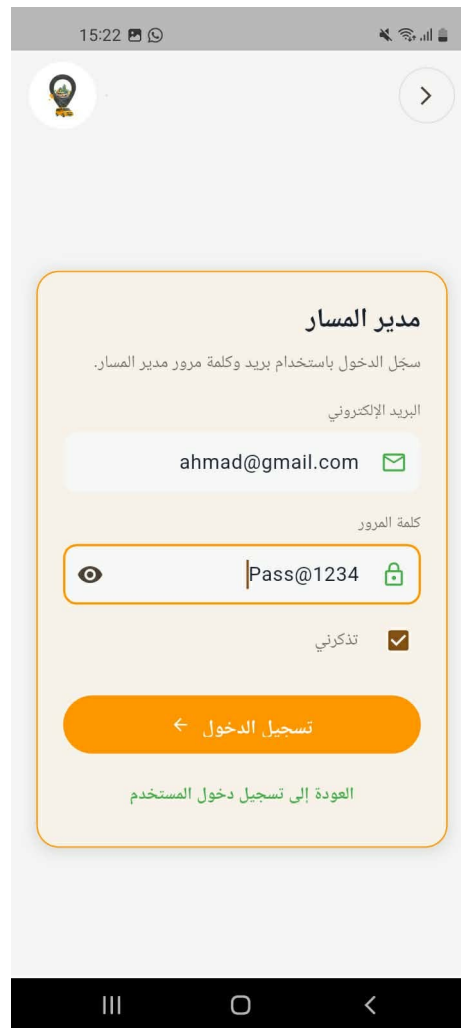


Figure 4.7: Route Manager login screen

4.1.8 Admin Login

Administrators access the web-based admin panel using their email address and password. The admin panel provides tools for managing users, routes, trips, reports, and driver approvals.

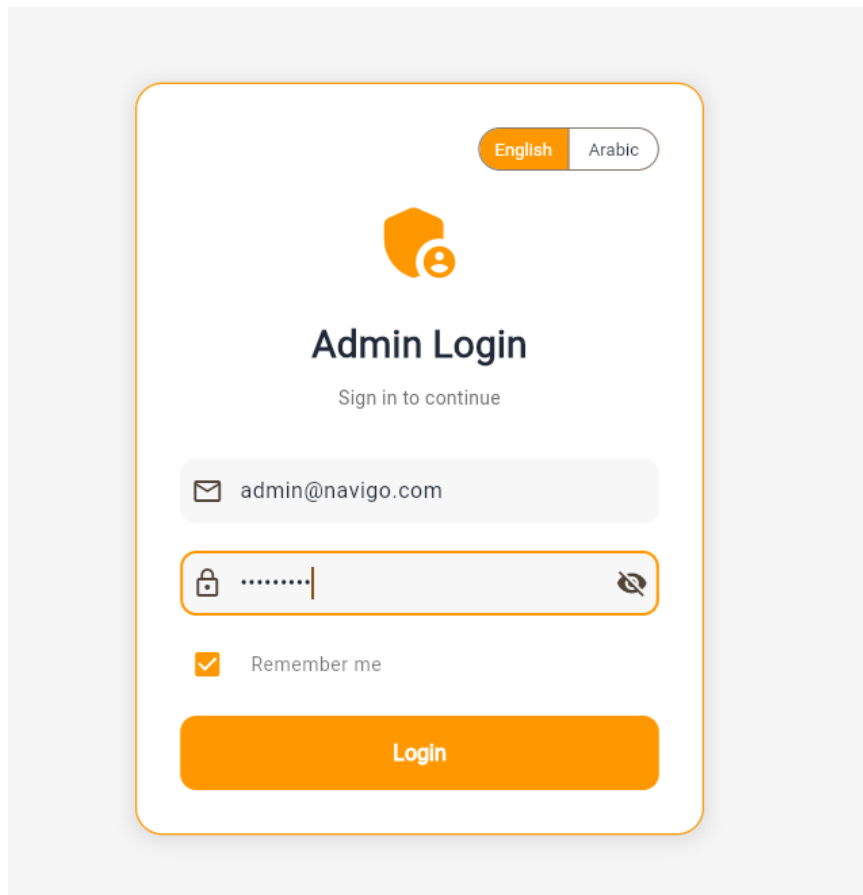


Figure 4.8: Administrator login interface

4.1.9 User Login

Registered users can log into the system using their phone number and OTP verification. This approach simplifies authentication while maintaining account security.

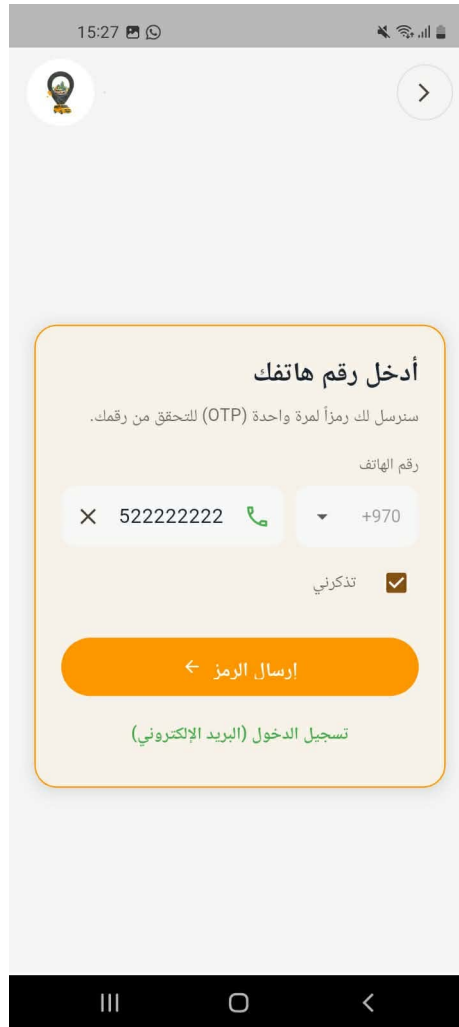


Figure 4.9: Registered-user login screen

4.2 Route Manager Module

The Route Manager module enables Route Managers to organize transportation operations efficiently. Through this module, they can create and manage trip schedules, monitor driver availability, review passenger and driver reports, manage waiting lists, and update their personal information.

4.2.1 Trip Schedule Management

The Route Manager can view all scheduled trips for the assigned route. Trips can be filtered by date and vehicle type, and new trips can be added or existing trips updated when necessary.

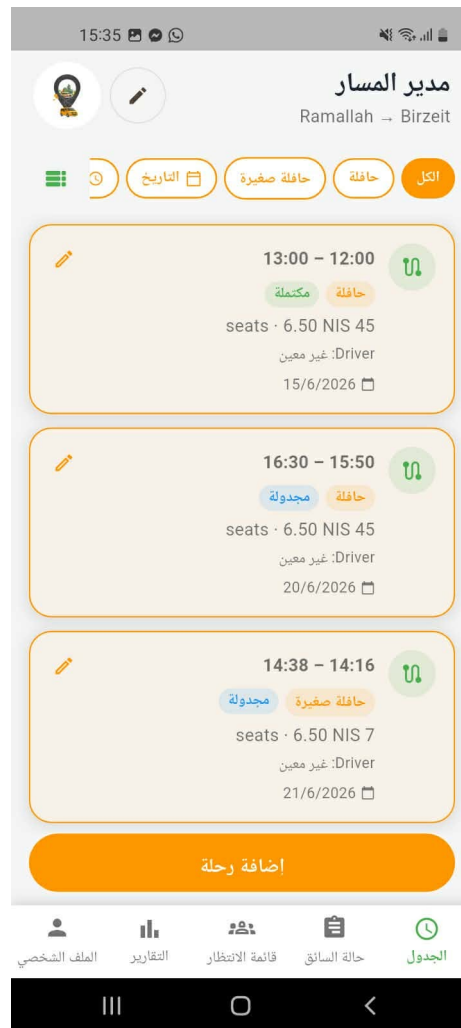


Figure 4.10: Route Manager trip-schedule management screen

4.2.2 Driver Status Monitoring

This screen allows Route Managers to monitor driver availability and connection status. Drivers can be filtered according to their current status to simplify trip assignment and operational planning.



Figure 4.11: Driver availability and connection-status monitoring

4.2.3 Create and Schedule Trips

Route Managers can create new trips by specifying departure and arrival times, the trip date, vehicle capacity, and ticket price. This helps ensure that transportation schedules are organized and available to passengers.

15:34

جدولة رحلة

نافذة المغادرة، التاريخ، والسعة

حافلة حافلة صغيرة

من 15:50 إلى 16:30

تكرار كل (دقائق، اختياري)

اتركه فارغاً لرحلة واحدة

التاريخ

20/6/2026

السعة (مقاعد)

seats 45

السعر

NIS 6.50

حفظ الرحلة

الملف الشخصي التقارير قائمة الانتظار حالة السائق الجدول

Figure 4.12: Trip creation and scheduling screen

4.2.4 Reports Management

The reports section allows Route Managers to review passenger and driver reports. Reports can be searched, reviewed, and forwarded to administrators when additional action is required.

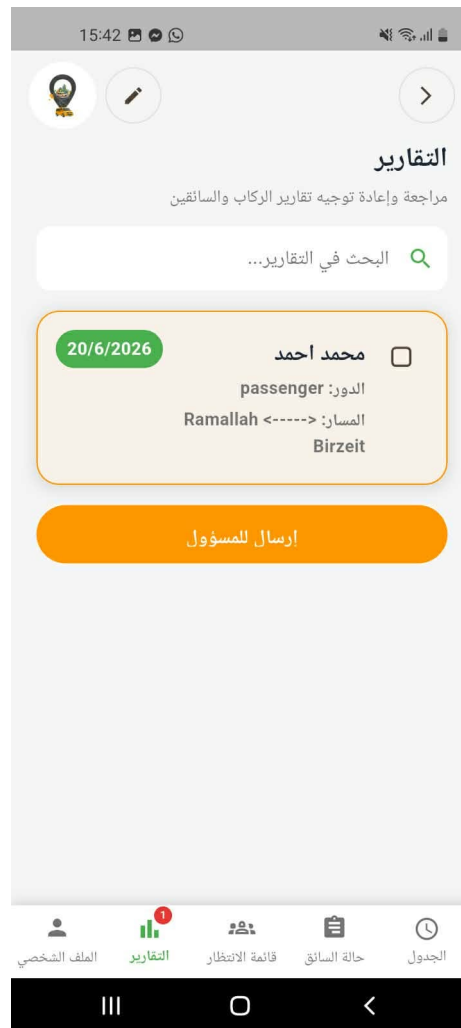


Figure 4.13: Passenger and driver reports-management screen

4.2.5 Waiting List Management

When no suitable trip is available, passengers may join a waiting list. Route Managers can review waiting-list requests and quickly create new trips based on passenger demand.

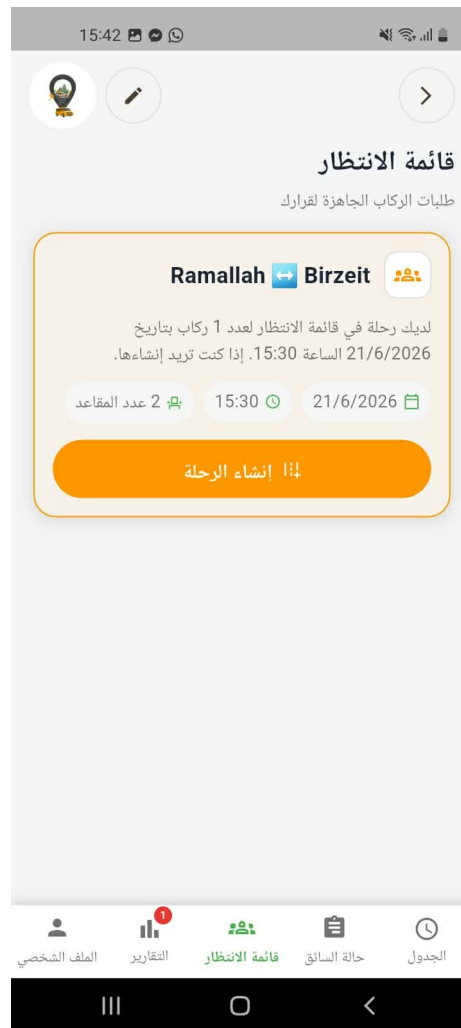


Figure 4.14: Passenger waiting-list management screen

4.2.6 Profile and Settings

The profile screen allows Route Managers to view and update their personal information, switch application language settings, change passwords, and securely log out of the system.

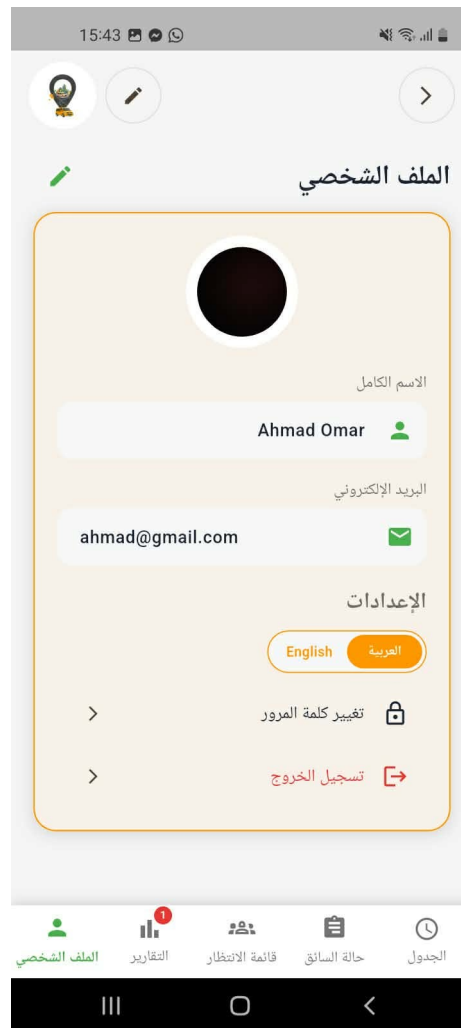


Figure 4.15: Route Manager profile and settings screen

4.3 Driver Module

The Driver module supports daily transportation operations by allowing drivers to manage assigned trips, receive passenger requests, track trip progress, view notifications, and update their profile information. These features help drivers communicate efficiently with passengers and Route Managers.

4.3.1 Driver Home Screen

The home screen provides drivers with an overview of their assigned trips and current route status. It also displays a map view that helps drivers monitor their location and access active-trip information quickly.

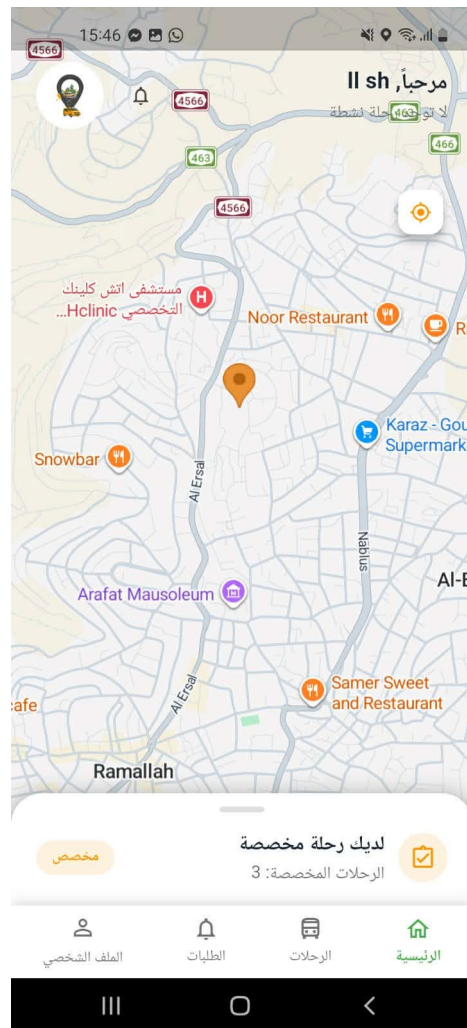


Figure 4.16: Driver home screen and route overview

4.3.2 Trip History and Status

Drivers can review all scheduled, active, and completed trips. Filtering options make it easier to track trip history and monitor the progress of ongoing transportation activities.

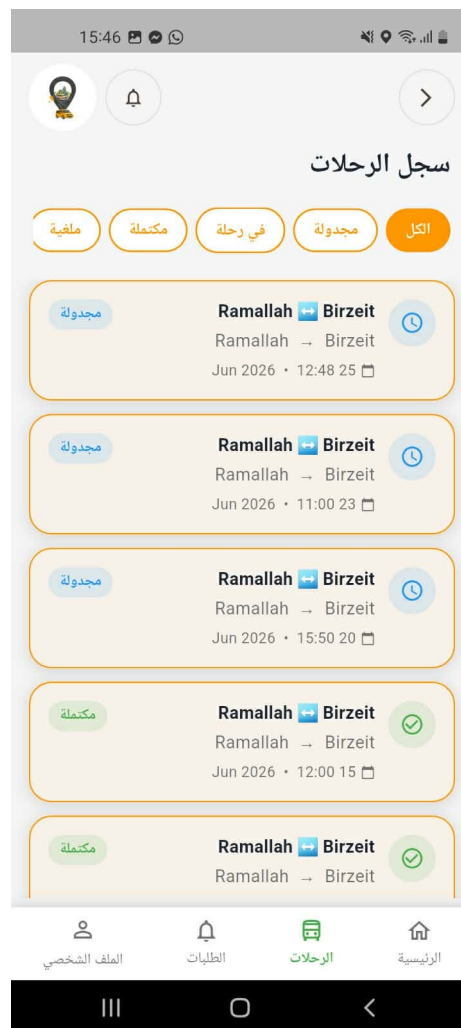


Figure 4.17: Driver trip history and status screen

4.3.3 Passenger Trip Requests

When passengers request seats on a trip, the request is displayed to the driver. The driver can review the request details and either accept or reject the booking.

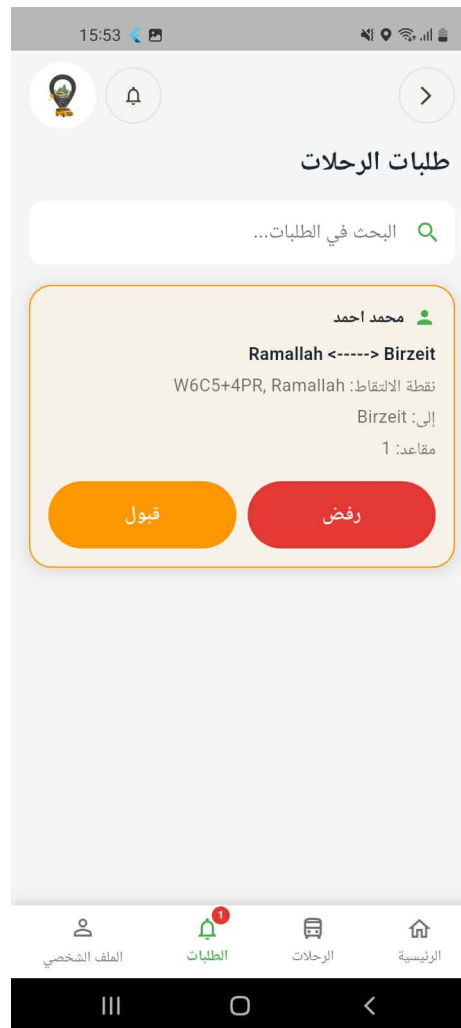


Figure 4.18: Passenger trip-request management screen

4.3.4 Driver Profile and Availability

The profile section allows drivers to view and update their personal information. Drivers can also change their availability status and manage application settings.

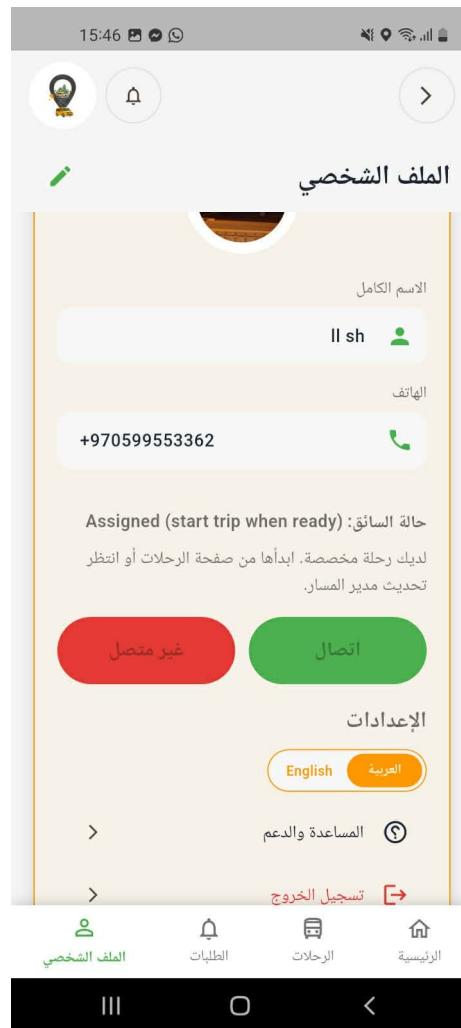


Figure 4.19: Driver profile and availability settings

4.3.5 Trip Details and Trip Start

Before starting a trip, drivers can review the route information, date, time, and assigned passengers. The trip can then be started or cancelled directly from this screen.

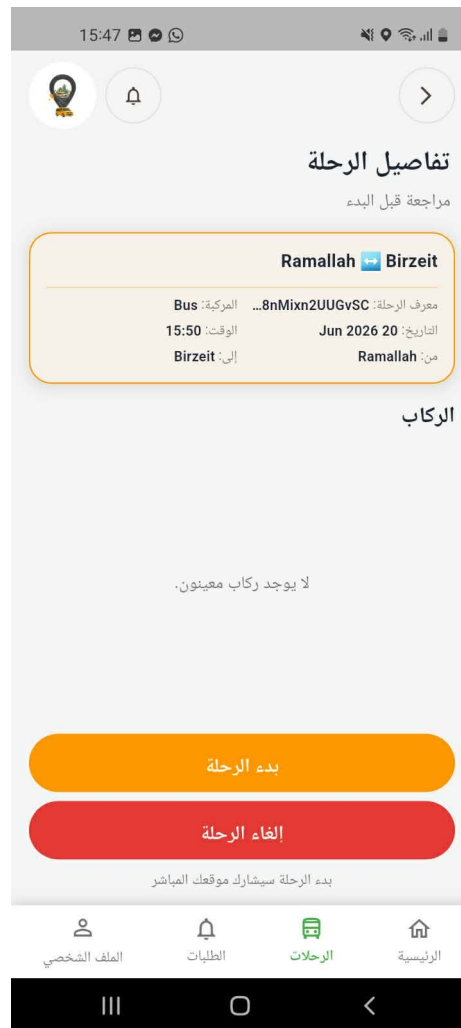


Figure 4.20: Trip details and trip-start controls

4.3.6 Live Trip Monitoring

During an active trip, drivers can monitor trip statistics such as capacity and passenger count. The system also displays the trip route on the map in real time.

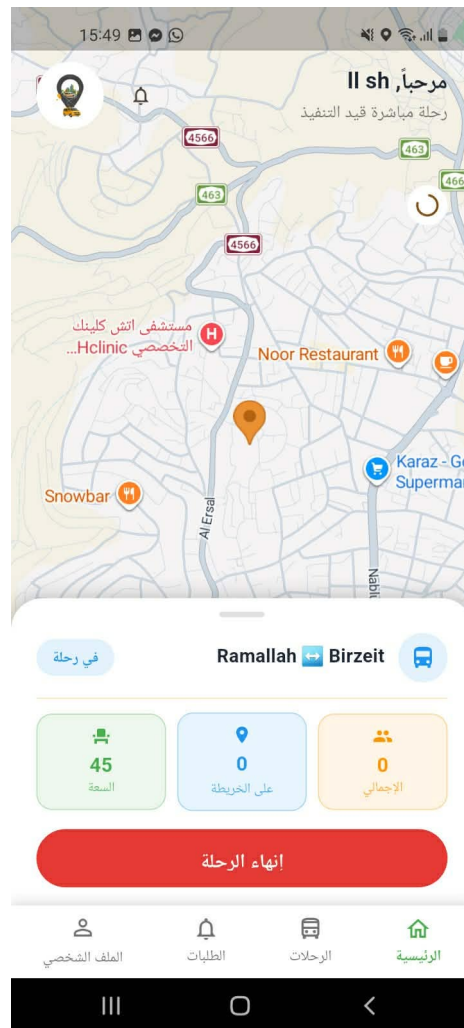


Figure 4.21: Live trip monitoring and route display

4.3.7 Notifications Center

The notifications screen informs drivers about new booking requests, trip updates, and other important events that require attention.



Figure 4.22: Driver notifications center

4.3.8 Push Notifications

Real-time push notifications are sent directly to the driver's device whenever a new passenger request or important trip update is received.

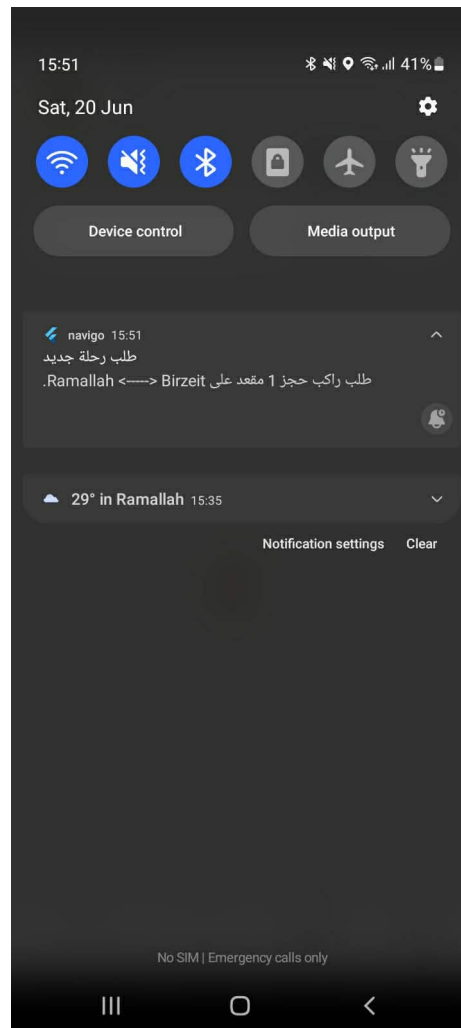


Figure 4.23: Real-time driver push notification

4.4 Passenger Module

The Passenger module enables users to search for available transportation routes, book seats, view trip details, track requests, receive notifications, manage personal information, and communicate with Route Management when support is needed. The module is designed to provide a simple and efficient booking experience.

4.4.1 Passenger Home Screen

The home screen displays the map interface, available transportation lines, and the passenger's current location. Users can search for routes and choose whether to book an immediate trip or schedule a future trip.

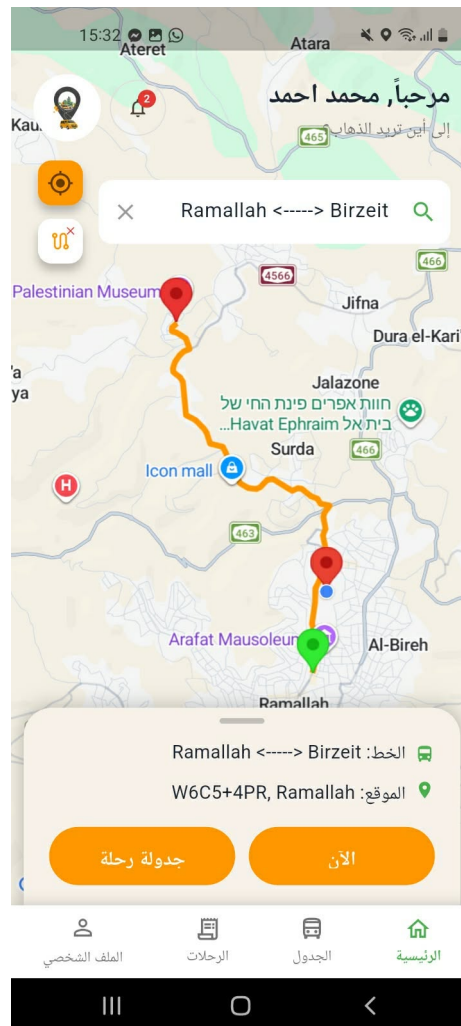


Figure 4.24: Passenger home screen and map interface

4.4.2 Trip Scheduling and Search

Passengers can filter available trips according to vehicle type, date, time, route, and required number of seats. The system then displays matching transportation options.

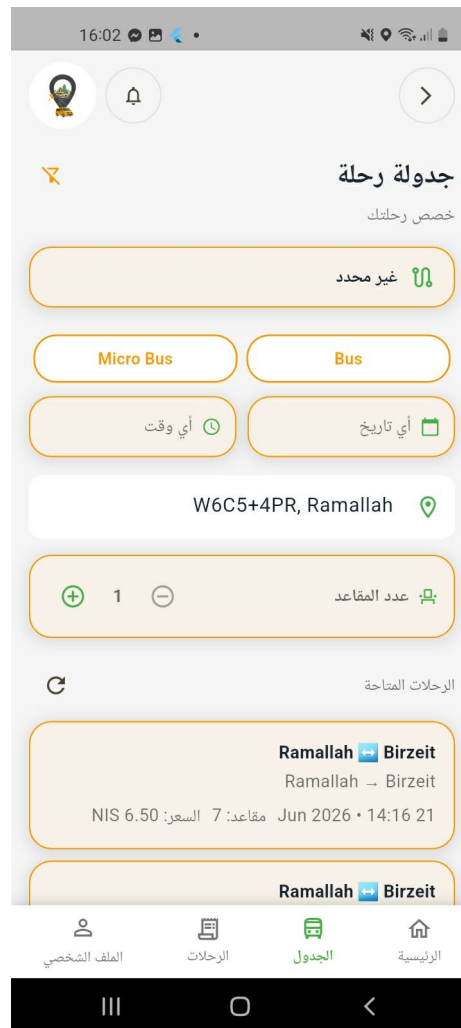


Figure 4.25: Passenger trip-scheduling and search filters

4.4.3 Trip Details

This screen provides complete information about a selected trip, including departure and arrival times, vehicle information, route details, driver contact information, and available seats.

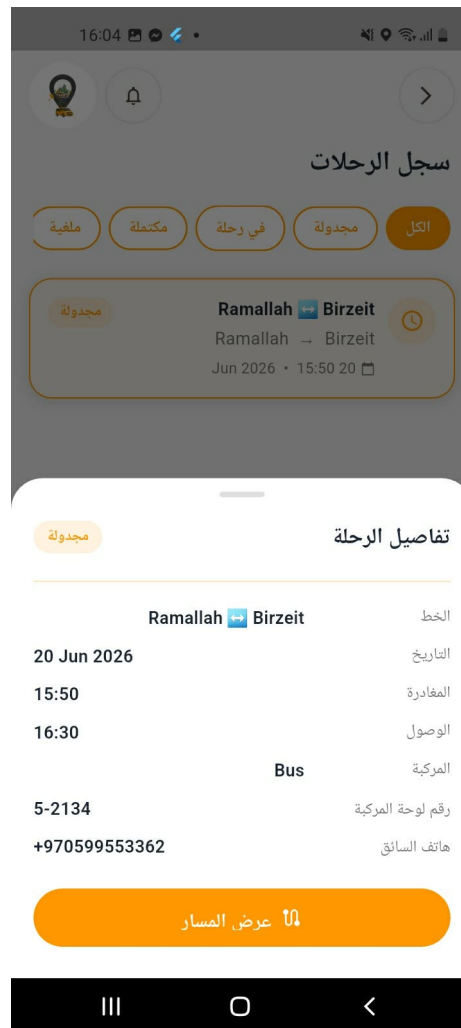


Figure 4.26: Selected-trip details screen

4.4.4 Passenger Profile

The profile screen allows passengers to view and manage their personal information, update preferences, switch the application language, and access account settings.

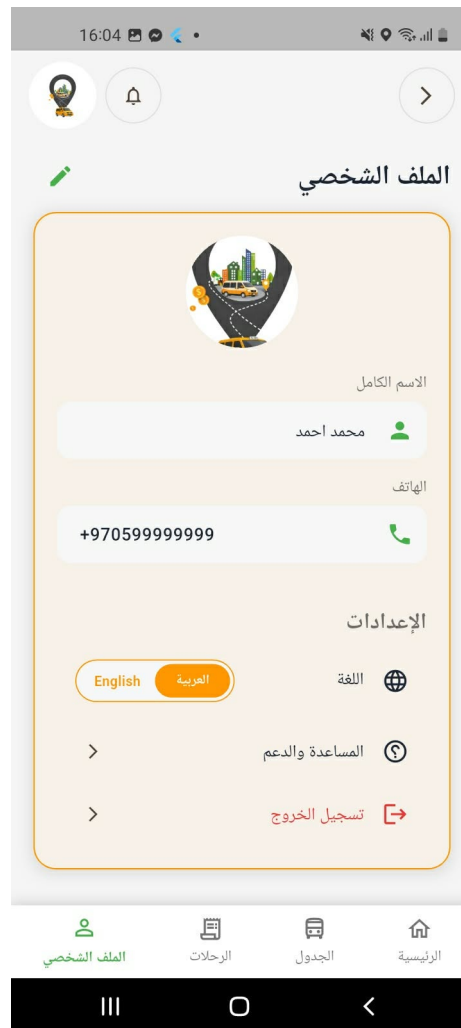


Figure 4.27: Passenger profile and account settings

4.4.5 Booking Confirmation Notification

After a booking request is accepted by the driver, the passenger receives a push notification confirming that the reservation has been approved successfully.

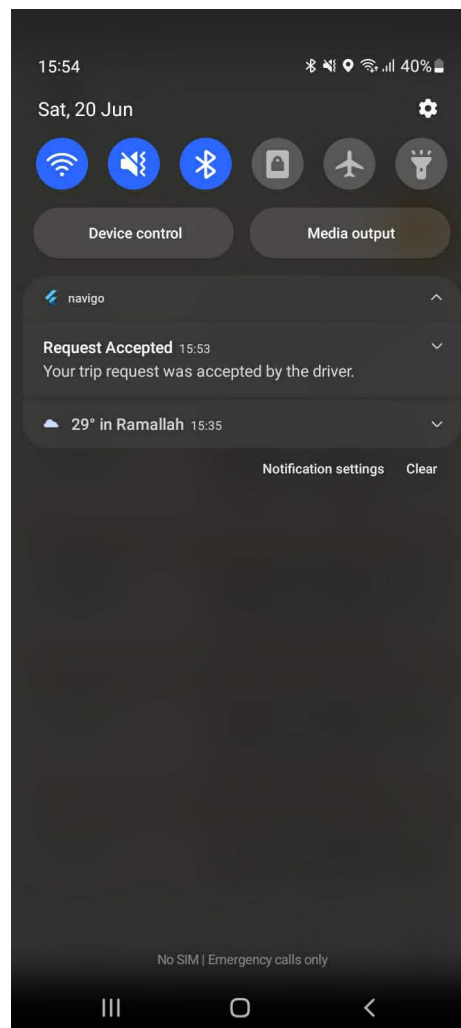


Figure 4.28: Passenger booking-confirmation notification

4.4.6 Route Search and Quick Booking

Passengers can quickly search for transportation lines directly from the map and immediately create a booking request for available trips.

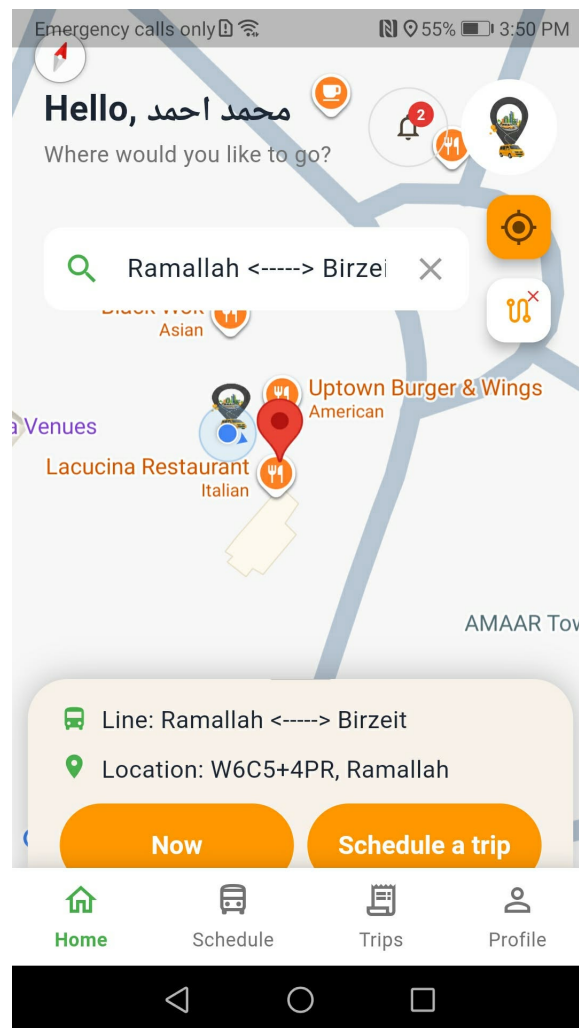


Figure 4.29: Route search and quick-booking interface

4.4.7 Trip Reservation Request

Before confirming a reservation, passengers can review vehicle details, route information, seat availability, and the driver's contact information. The number of requested seats can also be adjusted.

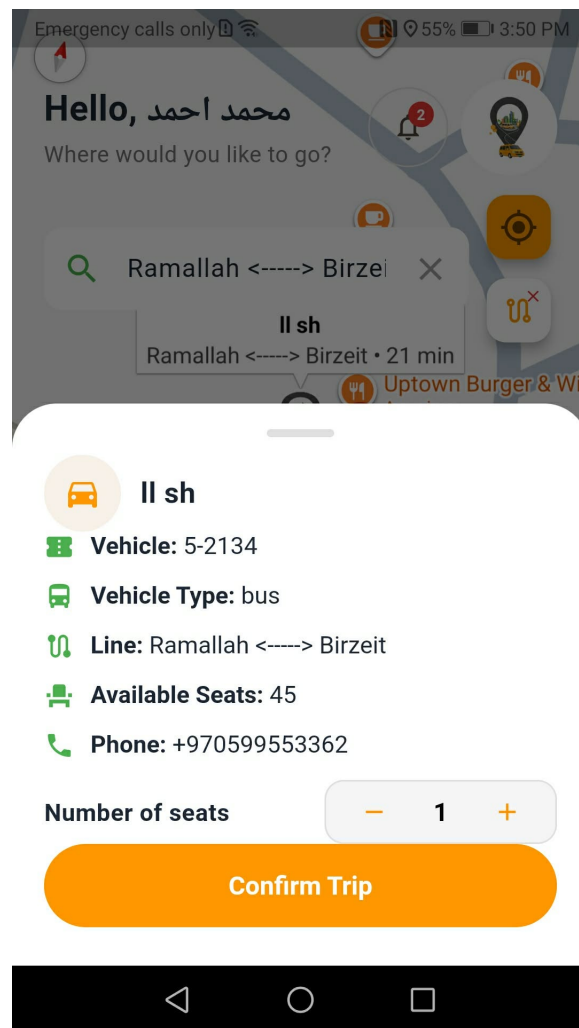


Figure 4.30: Passenger trip-reservation request screen

4.4.8 Support and Complaint Submission

The support screen enables passengers to submit complaints, feedback, or service-related issues directly to the Route Manager for review and resolution.

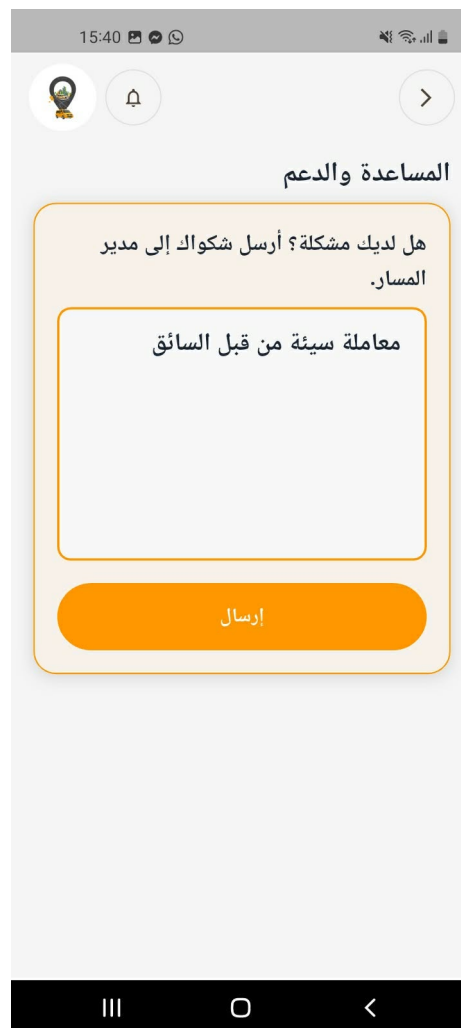


Figure 4.31: Passenger support and complaint-submission screen

4.5 Administrator Module

The Administrator module provides complete system-management capabilities. The administrator can monitor platform activity, manage routes and trips, view passenger information, approve drivers, analyze reports, and maintain system-wide settings through a centralized dashboard.

4.5.1 Administrator Dashboard

The dashboard provides a comprehensive overview of the system, including total users, active routes, active trips, pending driver approvals, and analytical summaries that support decision-making.

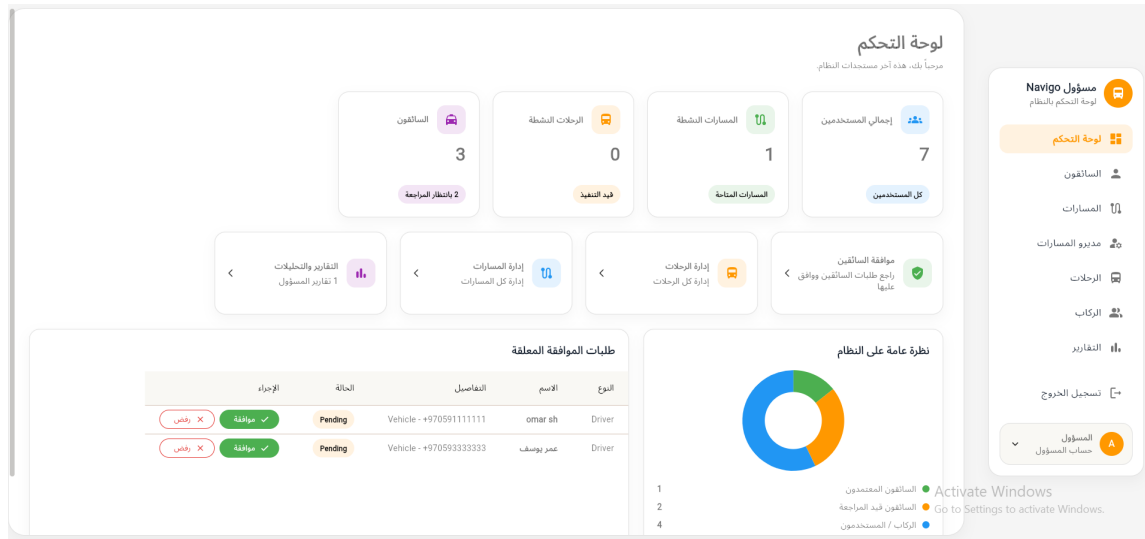


Figure 4.32: Administrator dashboard and system overview

4.5.2 Route Management and Route Creation

Administrators can create, update, and manage transportation routes. Route creation includes selecting start and destination points, defining coordinates, assigning vehicle types, and setting route pricing.

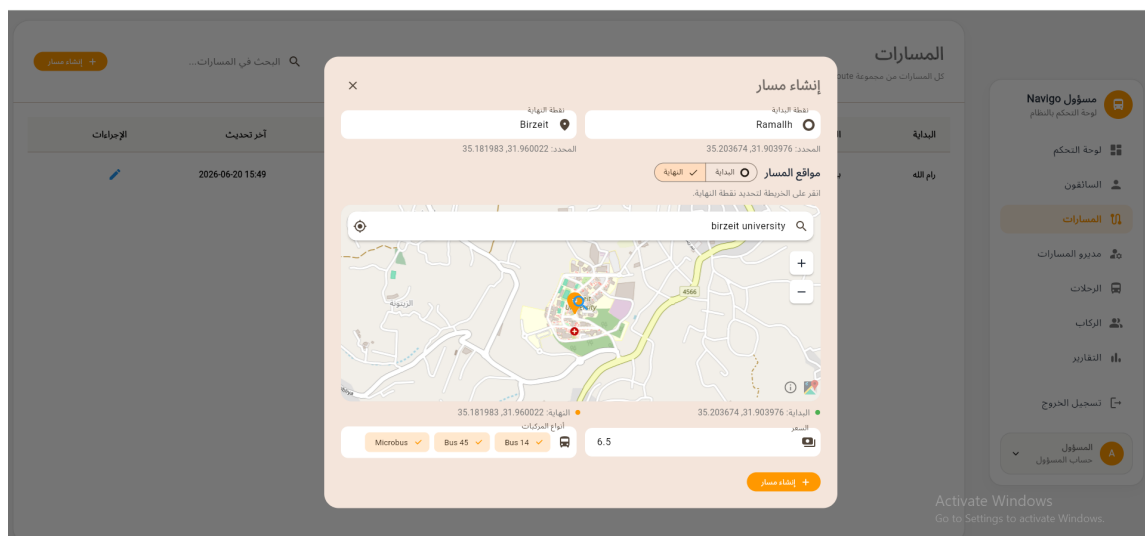


Figure 4.33: Administrator route-management and creation interface

4.5.3 Trip Management

This screen allows administrators to monitor all scheduled and active trips. Information such as route, vehicle type, departure time, arrival time, seat occupancy, and trip status is displayed for operational control.

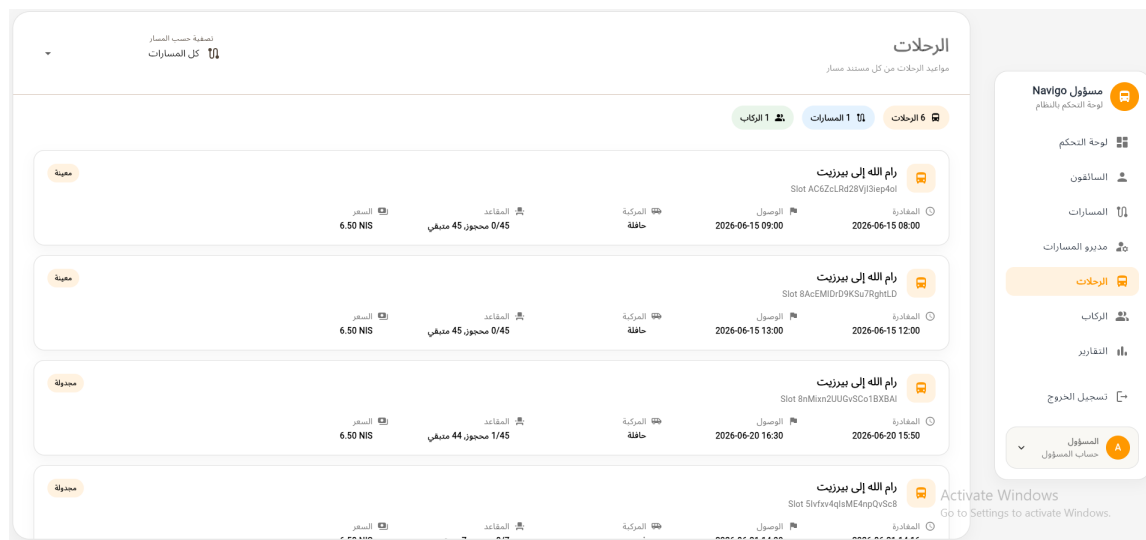


Figure 4.34: Administrator trip-management interface

4.5.4 Passenger Information Management

Administrators can review passenger account details, location information, connection status, activity history, and account metadata to support system monitoring and user management.

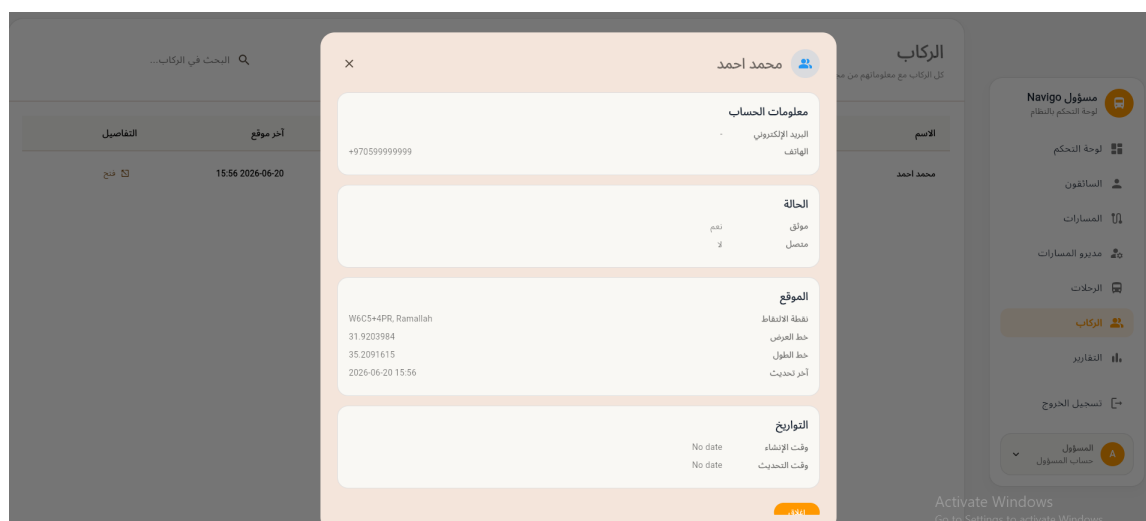


Figure 4.35: Passenger information-management interface

4.5.5 Account Security and Password Management

The administrator can securely update account credentials by verifying the current password and defining a new password to maintain account security.

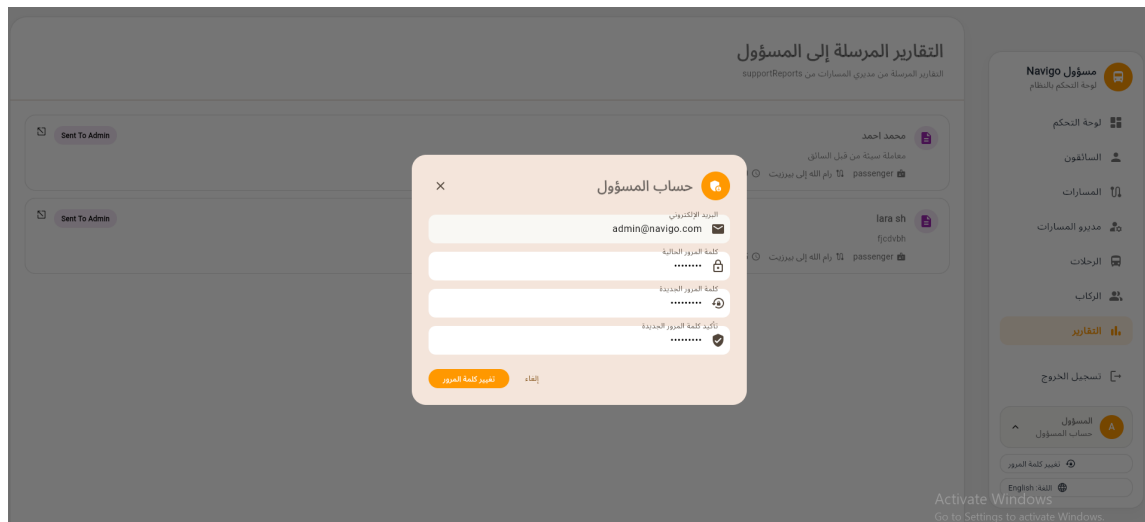


Figure 4.36: Administrator password-management interface

4.5.6 Reports and Complaint Management

Administrators receive reports submitted by passengers and Route Managers. Detailed report information includes sender details, route information, timestamps, status, and complaint content to facilitate issue resolution.

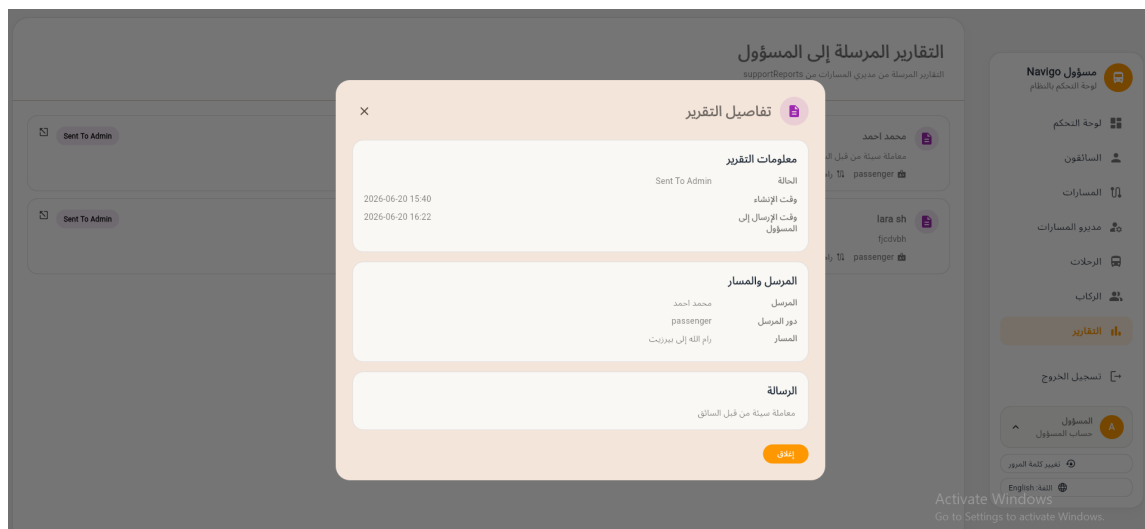


Figure 4.37: Reports and complaint-management interface

Chapter 5

Testing and Evaluation

To ensure that Navigo works correctly and meets user requirements, several test cases were performed on the system's main features. The testing process focused on validating core functionalities such as user registration, route browsing and trip requests, real-time tracking and notifications, support services, and admin operations.

Each test case was executed under specific conditions and compared against the expected results. The outcomes showed that the tested features functioned as intended and provided a reliable user experience across different user roles.

5.1 User Registration

This section presents the executed test cases for user registration and authentication in the Navigo application. The tests verify successful passenger registration, validation of duplicate phone numbers during driver registration, and Route Manager login using valid credentials.

5.1.1 Passenger Registration with a Valid Phone Number and OTP

Test date	20 June 2026
Test scenario	User Registration
Test designed by	Lara, Deema, and Cileen
Test description	Registering a new passenger account using valid information and a valid OTP code.
Preconditions	The user is not registered in the system; an Internet connection is available; and the user is on the Welcome Screen.

Table 5.1: Passenger registration test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Open the Navigo application.	The Welcome Screen appears.	Correct	Pass	None
2	Click <i>Get Started</i> .	Registration options appear.	Correct	Pass	None
3	Select Passenger as the user type.	The passenger registration form appears.	Correct	Pass	None
4	Enter the full name محمد احمد, phone number +970592222222, and required information.	The data is accepted.	Correct	Pass	None
5	Click <i>Register</i> .	The OTP verification screen appears.	Correct	Pass	None
6	Enter the OTP code 156804.	The OTP is verified successfully.	Correct	Pass	None
7	Complete the registration.	The home page is displayed.	Correct	Pass	Registration successful

Table 5.2: Passenger registration test execution

5.1.2 Driver Registration with an Already Registered Phone Number

Test date	20 June 2026
Test scenario	Driver Registration Validation
Test designed by	Lara, Deema, and Cileen
Test description	Attempting to register a new driver account using a phone number that is already associated with an existing account.
Preconditions	The phone number already exists in the Navigo system; the user is on the Registration Screen; and an Internet connection is available.

Table 5.3: Driver registration validation test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Open the Navigo application.	The Welcome Screen appears.	Correct	Pass	None
2	Click <i>Get Started</i> .	Registration options appear.	Correct	Pass	None
3	Select Driver as the user type.	The driver registration form appears.	Correct	Pass	None
4	Enter the name عمر يوسف, route Birzeit–Ramallah, plate number 5-1549, and vehicle type Bus (45 seats).	The data is accepted.	Correct	Pass	None
5	Enter the already registered phone number +970593333333.	The system checks the phone number's uniqueness.	Correct	Pass	None
6	Click <i>Register</i> .	An error appears: "This phone number is already registered. Please use another phone number."	Correct	Pass	Validation works correctly
7	Attempt to continue the registration.	Registration is blocked until a different phone number is entered.	Correct	Pass	Expected behavior

Table 5.4: Duplicate-phone driver registration test execution

5.1.3 Route Manager Login with Valid Credentials

Test date	20 June 2026
Test scenario	Route Manager Login
Test designed by	Lara, Deema, and Cileen
Test description	Logging in as a Route Manager using a valid email address and password.
Preconditions	A Route Manager account already exists, and an Internet connection is available.

Table 5.5: Route Manager login test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Open the Navigo application.	The Welcome Screen appears.	Correct	Pass	None
2	Click <i>Log In</i> .	Login options appear.	Correct	Pass	None
3	Select <i>Sign in By Email</i> .	The Route Manager login form appears.	Correct	Pass	None
4	Enter the email ahmad@gmail.com and password Pass@1234 .	The credentials are accepted.	Correct	Pass	None
5	Click <i>Login</i> .	The home page is displayed.	Correct	Pass	Login successful

Table 5.6: Route Manager login test execution

5.2 Route Browsing, Scheduling, and Trip Requests

This section presents the executed test cases for route browsing, trip scheduling and requests, waiting-list registration, and schedule management in the Navigo application.

5.2.1 Passenger Searches for and Selects a Route

Test date	20 June 2026
Test scenario	Route Browsing
Test designed by	Lara, Deema, and Cileen
Test description	The passenger searches for a route from the home screen and selects a route to display its path on the map.
Preconditions	The passenger is logged into the system; routes already exist in the database; and Internet connection and GPS services are enabled.

Table 5.7: Route browsing test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Open the Navigo application.	The home page appears.	Correct	Pass	None
2	Click the search box.	Available routes appear.	Correct	Pass	None
3	Enter the departure or destination location.	Matching routes are displayed.	Correct	Pass	None
4	Select the Birzeit–Ramallah route from the results.	Route details are displayed.	Correct	Pass	None
5	View the route on the map.	The route path is drawn on the map.	Correct	Pass	Map displayed successfully

Table 5.8: Route browsing test execution

5.2.2 Passenger Filters Trips and Requests a Trip

Test date	20 June 2026
Test scenario	Trip Filtering and Booking
Test designed by	Lara, Deema, and Cileen
Test description	The passenger filters available trips by date, time, and vehicle type, requests a trip, and joins the waiting list if no trips are available.
Preconditions	The passenger is logged in; routes exist in the system; and an Internet connection is available.

Table 5.9: Trip filtering and booking test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Open the <i>Schedule a Trip</i> page.	Route information appears.	Correct	Pass	None
2	Select the preferred time, date, and number of seats.	Trips are filtered using the entered data.	Correct	Pass	None
3	Choose an available trip on the Birzeit–Ramallah route.	Trip details are displayed.	Correct	Pass	None
4	Click <i>Request Trip</i> .	The trip request is created successfully.	Correct	Pass	None
5	If no trips are available, click <i>Join Waiting List</i> .	The waiting-list registration form appears.	Correct	Pass	None
6	Confirm the waiting-list registration.	The user is added to the waiting list successfully.	Correct	Pass	Waiting-list entry created

Table 5.10: Trip filtering and booking test execution

5.2.3 Route Manager Creates a Schedule Slot

Test date	20 June 2026
Test scenario	Schedule Management
Test designed by	Lara, Deema, and Cileen
Test description	The Route Manager creates a new schedule slot for an existing route.
Preconditions	The Route Manager is logged in using valid credentials; the route already exists; and an Internet connection is available.

Table 5.11: Schedule-slot creation test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Log in as the Route Manager using <code>ahmad@gmail.com</code> and password <code>Pass@1234</code> .	The home screen appears.	Correct	Pass	None
2	Open the <i>Add Schedule Slot</i> screen.	The routes list appears.	Correct	Pass	None
3	Select the Birzeit–Ramallah route.	The route details page opens.	Correct	Pass	None
4	Select vehicle type Bus (45 seats).	The selected vehicle type is accepted.	Correct	Pass	None
5	Enter the departure time, arrival time, capacity, and other information.	The data is accepted.	Correct	Pass	None
6	Save the schedule slot.	The schedule slot is created successfully.	Correct	Pass	None
7	Refresh the route page.	The new schedule slot is displayed.	Correct	Pass	Schedule created successfully

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
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Table 5.12: Schedule-slot creation test execution

5.3 Real-Time Tracking, Notifications, and Support

This section presents the executed test cases for live driver tracking, proximity and trip-status notifications, and support-report management in the Navigo application.

5.3.1 Driver Shares Live Location and Passenger Tracks Driver

Test date	20 June 2026
Test scenario	Real-Time Tracking
Test designed by	Lara, Deema, and Cileen
Test description	Verify that the driver shares a live location successfully, the passenger can track the driver on the map, and the passenger receives a notification when the driver is within 500 meters.
Preconditions	The driver and passenger are logged into the system; an active trip exists; and GPS and Internet connectivity are enabled.

Table 5.13: Real-time driver tracking test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	The driver logs into the application.	The driver dashboard appears.	Correct	Pass	None
2	The driver starts the trip and tracking.	The trip status changes to Active.	Correct	Pass	None
3	The passenger opens the trip details page.	Trip details are displayed.	Correct	Pass	None
4	The passenger selects <i>Track Driver</i> .	The driver's location appears on the map.	Correct	Pass	None
5	The driver moves toward the passenger's pickup location.	The passenger sees live movement on the map.	Correct	Pass	None

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
6	The driver reaches a distance of 500 meters or less.	The system sends the notification: “Your driver is nearby.”	Correct	Pass	Notification triggered successfully
7	The passenger opens the notification.	The driver-proximity message is displayed.	Correct	Pass	None
8	The driver continues moving.	The map updates the driver’s position in real time.	Correct	Pass	Tracking works successfully

Table 5.14: Real-time driver tracking test execution

5.3.2 Driver Approves Trip Request and Passenger Receives Notification

Test date	20 June 2026
Test scenario	Trip Approval and Notification
Test designed by	Lara, Deema, and Cileen
Test description	Verify that approved trips appear on the map, passengers can request a trip, drivers can approve the request, and the passenger receives a notification.
Preconditions	The driver is logged in and has approved trips available; the passenger is logged in; and an Internet connection is available.

Table 5.15: Trip approval and notification test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	The passenger opens the Home screen.	The home page appears.	Correct	Pass	None
2	The passenger clicks <i>Now</i> .	Available trips are displayed on the map.	Correct	Pass	None
3	The passenger selects a trip.	Trip details are displayed.	Correct	Pass	None

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
4	The passenger clicks <i>Request Trip</i> .	The trip request is sent to the driver.	Correct	Pass	None
5	The driver opens pending trip requests.	The passenger's request is displayed.	Correct	Pass	None
6	The driver clicks <i>Accept</i> .	The trip-request status changes to Accepted.	Correct	Pass	None
7	The system sends a notification to the passenger.	A trip-accepted notification is generated.	Correct	Pass	None
8	The passenger opens the notification.	The trip-acceptance message is displayed.	Correct	Pass	Notification received successfully

Table 5.16: Trip approval and notification test execution

5.3.3 Passenger or Driver Submits a Support Report and Route Manager Reviews It

Test date	20 June 2026
Test scenario	Support Management
Test designed by	Lara, Deema, and Cileen
Test description	Verify that a passenger or driver can submit a support report and that the Route Manager can review it successfully.
Preconditions	The user is logged in, and an Internet connection is available.

Table 5.17: Support management test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	The user opens the Support/Report section.	The support page appears.	Correct	Pass	None

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
2	The user enters the report details.	The data is accepted.	Correct	Pass	None
3	The user clicks <i>Submit Report</i> .	The report is stored successfully.	Correct	Pass	None
4	The Route Manager logs into the report screen using <code>ahmad@gmail.com</code> and password <code>Pass@1234</code> .	The Route Manager dashboard appears.	Correct	Pass	None
5	The Route Manager reviews the report.	The report status is updated successfully.	Correct	Pass	Support workflow completed
6	The Route Manager selects the submitted report and sends it to the admin.	The report details are displayed.	Correct	Pass	None
7	The admin opens <i>Support Reports</i> .	The reports list is displayed.	Correct	Pass	None

Table 5.18: Support management test execution

5.4 Admin Management

This section presents the executed test cases for driver-account approval and route management through the Navigo admin panel.

5.4.1 Admin Approves a Pending Driver Account

Test date	20 June 2026
Test scenario	Driver Approval Management
Test designed by	Lara, Deema, and Cileen
Test description	Verify that the admin can review a pending driver account and approve it successfully from the admin dashboard.
Preconditions	The admin is logged into the admin panel; at least one driver account is pending approval; and an Internet connection is available.

Table 5.19: Pending-driver approval test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Log into the admin panel using <code>admin@navigo.com</code> and password <code>Pass1234</code> .	The admin dashboard appears.	Correct	Pass	None
2	Go to the <i>Drivers Approval</i> section.	The pending-drivers list is displayed.	Correct	Pass	None
3	Select the pending driver account for <code>عمر يوسف</code> .	The driver details are displayed, including route Birzeit–Ramallah, phone number +970593333333, plate number 5-1549, and vehicle type Bus (45 seats).	Correct	Pass	None
4	Click <i>Approve</i> .	The confirmation action is processed.	Correct	Pass	None
5	Automatically refresh the drivers list.	The driver status changes from Pending to Approved.	Correct	Pass	Driver approved successfully

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
6	Check the system dashboard.	The pending-drivers count is updated.	Correct	Pass	Dashboard updated correctly

Table 5.20: Pending-driver approval test execution

5.4.2 Admin Rejects a Pending Driver Account

Test date	20 June 2026
Test scenario	Driver Approval Management
Test designed by	Lara, Deema, and Cileen
Test description	Verify that the admin can reject a pending driver account that does not meet the approval requirements.
Preconditions	The admin is logged into the admin panel; at least one driver account is pending approval; and an Internet connection is available.

Table 5.21: Pending-driver rejection test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Log into the admin panel using <code>admin@navigo.com</code> and password <code>Pass1234</code> .	The admin dashboard appears.	Correct	Pass	None
2	Go to the <i>Drivers Approval</i> section.	The pending-drivers list is displayed.	Correct	Pass	None
3	Select a pending driver account.	The driver details are displayed.	Correct	Pass	None
4	Click <i>Reject</i> .	The driver-rejection action is processed.	Correct	Pass	None

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
5	Automatically refresh the drivers list.	The rejected driver is removed from pending approvals or marked as rejected.	Correct	Pass	Driver rejected successfully
6	Check the system dashboard.	The pending-drivers count is updated.	Correct	Pass	Dashboard updated correctly

Table 5.22: Pending-driver rejection test execution

5.4.3 Admin Creates a New Route

Test date	20 June 2026
Test scenario	Route Management
Test designed by	Lara, Deema, and Cileen
Test description	Verify that the admin can create a new route by entering route details, selecting start and end points on the map, choosing vehicle types, and saving the route.
Preconditions	The admin is logged into the admin panel and is on the Routes Management page; map service and Internet connectivity are available.

Table 5.23: Route creation test-case information

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
1	Log into the admin panel using <code>admin@navigo.com</code> and password <code>Pass1234</code> .	The admin dashboard appears.	Correct	Pass	None
2	Go to <i>Routes Management</i> .	The Routes Management page is displayed.	Correct	Pass	None
3	Click <i>Create Route</i> .	The Create Route form appears.	Correct	Pass	None

ID	Step Description	Expected Result	Actual Result	Status	Additional Notes
4	Enter the start point Birzeit and end point Ramallah.	The route locations are accepted.	Correct	Pass	None
5	Select the start and end locations on the map.	Start and end markers are set on the map.	Correct	Pass	None
6	Enter the route price.	The price field accepts the value.	Correct	Pass	None
7	Select vehicle type Bus (45 seats).	The selected vehicle type is highlighted.	Correct	Pass	None
8	Click <i>Create Route</i> .	The route is created and saved successfully.	Correct	Pass	Route added successfully
9	Return to the routes list.	The new Birzeit–Ramallah route appears in the routes list.	Correct	Pass	Route creation completed

Table 5.24: Route creation test execution

5.5 Testing Summary

Overall, all defined functional test cases were executed successfully. The tested scenarios covered user registration, OTP verification, route browsing, trip scheduling, seat availability calculation, trip requests, real-time tracking, notifications, support reports, and admin/route manager workflows. The results confirm that the main system modules operate correctly under the tested conditions.

Chapter 6

Conclusion and Future Work

6.1 Review of the Project

This project presented the analysis, design, implementation, and testing of the Navigo-^{وَصَلِّي} smart transportation system. The work focused on understanding the transportation problem, defining clear functional and non-functional requirements, modeling the system using standard software engineering techniques, and implementing the main system modules.

Throughout the project, key system components were identified, including user roles, trip management, route browsing, real-time tracking, cash payment handling, and notification services. UML models such as use case diagrams, class diagrams, sequence diagrams, activity diagrams, and state chart diagrams were used to clearly describe system behavior, structure, and interactions.

After completing the analysis and design phase, the Navigo-^{وَصَلِّي} system was implemented and evaluated through functional testing. Overall, the project achieved its objectives by delivering a working application that addresses real transportation needs through passenger, driver, route manager, and administrator workflows.

6.2 Project Completion Summary

The final stage of the Navigo-^{وَصَلِّي} project completed the implementation, integration, testing, and documentation activities. The implemented system includes passenger, driver, route manager, and administrator modules, with Firebase authentication, OTP verification, real-time tracking, seat availability calculation, cash payment handling, and functional testing of the main workflows.

Stage	Completed Activities	Period
1	Kickoff meeting and planning development phases; preparing sprint backlog and task breakdown.	February 2026
2	Implemented core backend logic including Firebase structure, authentication, and trip model.	Mid Feb – 01 Mar, 2026
3	Completed passenger mobile application development, including UI, maps, registration, seat availability, and real-time updates.	01 Mar – 20 Mar, 2026
4	Completed driver mobile application development, including trip management, availability status, and live tracking.	20 Mar – 10 Apr, 2026
5	Integrated tracking services, route management, notifications, and cash payment status.	April 10 – April 25, 2026
6	Executed functional testing for the main implemented workflows.	April 25 – May 5, 2026
7	Prepared the final project report draft and submitted it for supervisor review.	May 5 – May 20, 2026
8	Completed final corrections, documentation polishing, and presentation preparation.	May 20 – May 30, 2026
9	Completed final submission and project presentation.	June 2026

Table 6.1: Completed activities for the Navigo-وصلي project

6.3 System Limitations

Although the Navigo-وصلي system implements the main required functions, it has some practical limitations that should be considered during real use and future deployment:

- The system requires internet connectivity for real-time features such as tracking, notifications, and data synchronization.
- GPS accuracy may vary depending on device quality, signal strength, and the user's location.
- OTP verification depends on the availability of Firebase and SMS services.
- Real deployment requires cooperation with licensed drivers and transportation offices.
- Seat availability depends on correct trip and seat information entered into the system.
- Large-scale deployment may require further performance evaluation.

6.4 Future Work

1. Digital Payment Integration: Add online payment options such as Jawwal Pay in a future version. This enhancement would require secure payment gateway integration, transaction confirmation, refund handling, and appropriate financial security

controls.

2. **AI-Based Route and Demand Prediction:** Use artificial intelligence to analyze historical trip data in order to predict peak hours, suggest optimal routes, and improve vehicle allocation. This will help reduce congestion and enhance overall service efficiency.
3. **Door Camera Passenger Counting:** Install an entry/exit camera at the vehicle door to automatically count boarding and alighting passengers and update vehicle occupancy in real time. This replaces the current manual driver updates, reduces errors and delays, enables capacity alerts, and generates accurate passenger-flow analytics while preserving privacy.
4. **Smart Driver Distribution Using Heat Maps:** Develop an intelligent driver allocation system that analyzes real-time demand and historical usage to generate heat maps and recommend where drivers should move or wait. This helps reduce congestion, improve coverage across the entire route, and minimize passenger waiting time.
5. **Integrate QR Code devices on buses** to enable passengers to scan for seamless boarding, trip validation, and instant notifications, enhancing the overall user experience and operational efficiency.

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Appendix A

Appendices

A.1 Use Case Specifications

This section provides detailed specifications for the main use cases of the Navigo – وصلني system. Each use case is presented in a structured format including actors, description, conditions, main flow, alternative flows, exception flows, and postconditions.

A.1.1 UC4: Realtime Tracking

Item	Description
Use Case Name	Realtime Tracking
Actors	Passenger: Tracks the live location and ETA of assigned vehicles. Driver: Updates and shares live location during trips. Route Manager: Monitors vehicle locations for operational management. Admin: Oversees tracking system performance and data accuracy.
Description	This use case allows passengers to monitor the real-time location and estimated arrival time (ETA) of their assigned vehicle during a trip. Drivers continuously update their location, enabling accurate tracking. Route managers and admins use tracking data to support efficient operations and timely service delivery.
Preconditions	Passenger and driver are logged into the system. GPS and map services are enabled and functioning. A trip is active and assigned to the passenger and driver. Network connectivity is available for data transmission.
Trigger	Passenger opens the trip tracking screen. Driver starts a trip and enables location sharing. The system periodically updates vehicle location data.

Item	Description
Main Flow	1. Passenger opens the “Trip Tracking” screen. 2. The system retrieves the live location of the assigned vehicle. 3. The system displays the vehicle’s real-time position on the map. 4. The system calculates and continuously updates ETA. 5. Passenger monitors vehicle movement and ETA until pickup or trip completion. 6. Driver’s device continuously sends location updates to the system. 7. Route manager and admin monitor tracking dashboards for operational insights. 8. Upon trip completion, the tracking session ends and data is archived.
Alternative Flow	A1 – Driver stops sharing location: The driver loses GPS signal or disables location sharing; the system notifies the passenger, and the passenger may request support or manual updates. A2 – Passenger views past trip history: Passenger selects a completed trip; the system displays the route taken with timestamps. A3 – Fleet overview for admin/manager: Admin or route manager opens the live dashboard; the system displays locations and statuses of active vehicles, and alerts/rerouting actions may be applied.
Exception Flow	E1 – No GPS signal or service unavailable: The system displays “Unable to retrieve live location. Please check your connection or try again later.” E2 – Network connectivity loss: Location updates are suspended temporarily; the system attempts reconnection and resumes automatically. E3 – Unauthorized access: The system denies access to users not linked to the trip and displays “You do not have permission to view this trip’s location.”
Data	Trip ID, passenger ID, driver ID, vehicle ID, GPS coordinates (latitude, longitude), timestamp of location update, estimated arrival time (ETA), trip status.
Postconditions	Passenger successfully monitors the assigned vehicle’s live location and ETA. Driver location is continuously updated and visible only to authorized users. Tracking data is stored for future reference and analytics. Any tracking disruption is communicated to the passenger.

Item	Description
Comments	Accurate ETA depends on real-time GPS data combined with map routing. Supports all vehicle types (bus, microbus, taxi, etc.). Privacy controls ensure tracking data is accessible only to authorized users. System performance depends on reliable GPS and network availability.

Table A.1: Use Case Specification: Realtime Tracking

A.1.2 UC5: Trip Confirmation and History

Item	Description
Use Case Name	Trip Confirmation and History
Actors	Passenger: Books trips, receives trip confirmations, views trip history, and receives live updates on trip status. Driver: Updates trip statuses (arrived, started, completed) and views assigned trips. Admin: Monitors trip records, reviews trip history, and manages system-wide trip data. Notification Service: Sends trip confirmations and status updates to passengers and drivers.
Description	This use case manages trip booking confirmation and trip history storage. After a passenger books a trip, the system sends a confirmation notification including driver details, vehicle type, estimated fare, and pickup time. Users can view their complete trip history, including completed, canceled, and scheduled trips. Drivers update trip statuses, and passengers receive real-time notifications, ensuring transparency and accurate tracking of trip activities.
Preconditions	Passenger or driver is logged into the system. The trip has been successfully booked. Notification service is available for sending confirmations and updates.
Trigger	Passenger completes a trip booking or driver updates the status of an active trip.
Main Flow	1. Passenger successfully books a trip. 2. The system generates trip confirmation details (driver information, fare estimate, pickup time, and vehicle type). 3. The system stores the trip in the passenger's trip history. 4. Notification service sends a trip confirmation to the passenger. 5. The driver receives the assigned trip in their dashboard. 6. The driver updates the trip status (arrived, started, completed). 7. The system logs all status changes in trip history. 8. Passengers receive notifications for each status update.
Alternative Flow	A1 – Passenger views trip history: The passenger opens the Trip History section, the system retrieves all past trips (completed, canceled, scheduled), and displays their details. A2 – Admin reviews trip records: The admin opens the trip management dashboard, views all trips across the platform, and filters or reviews trips for reports, issues, or support cases.

Item	Description
Exception Flow	E1 – Notification service unavailable: The system stores the trip confirmation but displays “Trip confirmed, but notifications cannot be delivered right now.” E2 – Trip status update failed: The driver attempts to update trip status while the system is offline; the system displays “Unable to update trip status. Please try again.” E3 – Trip history not loading: The system fails to retrieve trip history and displays “Unable to load trip history. Please refresh or try again later.”
Data	Passenger ID, driver ID, trip ID, trip status (Scheduled, Arrived, Started, Completed, Canceled), fare estimate, vehicle details, timestamps and duration, trip location data.
Postconditions	Trip is confirmed and stored in the system history. Passengers and drivers receive notifications for trip updates. Trip records remain accessible for future review, reporting, and customer support.
Comments	Trip status updates must occur in real time. Trip history must be stored permanently to meet security and reporting requirements.

Table A.2: Use Case Specification: Trip Confirmation and History

A.1.3 UC6: Notifications and Live Updates

Item	Description
Use Case Name	Notifications and Live Updates
Actors	Passenger: Receives notifications about trip acceptance, driver arrival, delays, updated ETA, cancellations, and payment status; may disable push notifications which triggers SMS fallback; acts on notifications (e.g., prepares for pickup). Driver: Receives notifications about new trip requests, updates, cancellations, and schedule changes; sends trip status updates (Arrived, On Trip, Delayed) that trigger passenger notifications. Notification Service (Push): Sends real-time push notifications, handles delivery, retries, and failure detection, and manages device tokens. SMS Service: Sends SMS fallback messages when push notifications are disabled or fail to ensure critical alerts reach users. GPS Service: Provides real-time driver location updates, triggers location-based notifications, and supports ETA recalculation.
Description	The system delivers real-time notifications to passengers and drivers regarding trip acceptance, driver arrival, delays, updated ETA, cancellations, payment confirmation, and safety alerts. When push notifications are unavailable or disabled, SMS fallback is used for critical notifications to ensure timely communication.
Preconditions	User is authenticated and logged in. Internet connectivity is available for both passenger and driver. Notification and SMS services are operational.
Trigger	Any update in trip status, driver movement, route timing, or system event.
Main Flow	1. Passenger submits a trip request. 2. Driver accepts the trip and the system sends a push notification to the passenger. 3. GPS updates the driver's location. 4. As the driver approaches the pickup point, the passenger receives a "Driver is arriving" notification. 5. If delays or traffic occur, the system recalculates ETA. 6. The system sends an updated ETA notification. 7. Drivers receive notifications for new trips, changes, or cancellations.

Item	Description
Alternative Flow	A1 – Push notifications disabled: Passenger disables push notifications; the system activates SMS fallback for verification and critical trip updates. A2 – Manual trip status update: Driver updates the trip status (Arrived, On Trip, Delayed); the system pushes corresponding updates to the passenger.
Exception Flow	E1 – Notification service failure: The system retries sending notifications up to three times. E2 – No internet connection: Notifications are queued and delivered when connectivity is restored. E3 – Weak GPS signal: The system displays the last known driver location with a “GPS Weak” alert.
Data	User ID, driver ID, GPS coordinates, ETA data, trip status.
Postconditions	Notifications are successfully delivered to users or queued for later delivery.
Comments	This use case relies heavily on GPS accuracy and the availability of notification services.

Table A.3: Use Case Specification: Notifications and Live Updates

A.1.4 UC7: Cash Payment Handling

Item	Description
Use Case Name	Cash Payment Handling
Actors	Passenger: Confirms cash payment for a trip. Driver: Confirms that the cash fare has been collected. Notification Service: Sends payment status notifications to passengers and drivers.
Description	This use case allows passengers to complete trip payment using cash. The system displays the estimated fare, marks the payment as cash pending, and allows the driver to confirm payment collection after the passenger boards or the trip is completed.
Preconditions	Passenger is logged in. Passenger has an active booking or is completing a new reservation. The trip fare has been calculated by the system.
Trigger	Passenger confirms the trip with cash payment.
Main Flow	1. Passenger selects or confirms cash payment. 2. The system displays the total fare and payment details. 3. Passenger confirms the trip request or reservation. 4. The system marks the payment status as “Cash Pending”. 5. The system sends the reservation or trip confirmation to the passenger. 6. The system notifies the driver that cash will be collected. 7. After collecting the fare, the driver confirms payment collection. 8. The system updates the payment status to “Cash Collected”.
Alternative Flow	A1 – Fare updated before confirmation: The system displays the updated fare and asks the passenger to confirm the trip again. A2 – Driver marks payment as not collected: The system keeps the payment status as “Cash Pending” and records the issue for follow-up.
Exception Flow	E1 – Fare cannot be calculated: The system displays “Fare unavailable. Please try again later.” E2 – Driver confirmation fails: The system keeps the previous payment status and displays “Unable to update payment status. Please try again.”
Data	Trip ID, passenger ID, driver ID, amount, cash payment status, collection confirmation time.
Postconditions	Trip payment is marked as cash pending or cash collected. Payment status updates are logged. Payment notifications are sent to relevant actors.

Item	Description
Comments	Digital payment methods such as Jawwal Pay are outside the current system scope and are listed as future work. Cash payment status should be clear to both the passenger and the driver.

Table A.4: Use Case Specification: Cash Payment Handling

A.1.5 UC8: Route and Vehicle Management (Admin Panel)

Item	Description
Use Case Name	Route and Vehicle Management
Actors	Route Manager: Manages vehicles and schedules; assigns drivers using the queue system. Admin: Overrides or rejects modifications, deactivates drivers, manages system-level settings, and enforces authorization and permissions.
Description	This use case enables the Route Manager to manage transportation routes, vehicles, schedules, pricing, and driver availability. Drivers are assigned to routes using a queue-based mechanism; when one driver queue reaches capacity, the system automatically allocates subsequent assignments to other available drivers. The Admin oversees and approves changes when required to ensure system integrity.
Preconditions	Route Manager is logged into the system. Role-based permissions are enforced.
Trigger	Route Manager selects “Manage Routes” or “Manage Vehicles” from the administrative dashboard.
Main Flow	1. Route Manager logs into the dashboard. 2. Opens the “Routes & Vehicles Management” section. 3. The system displays all routes, pricing, schedules, and driver information. 4. Route Manager assigns drivers using the driver queue system. 5. The system saves changes and updates the passenger application in real time.
Alternative Flow	A1 – Admin deactivates a driver: The system automatically updates route availability. A2 – Schedule-only update: The Route Manager updates schedule times while routes remain unchanged.
Exception Flow	E1 – Invalid input: Missing required fields trigger a validation error. E2 – Unauthorized access: The system displays “Access Denied.” E3 – Modification conflict: The system locks the record to prevent simultaneous edits.
Data	Route ID, vehicle details, driver profiles, schedule times, pricing data.
Postconditions	Updated route and vehicle information is stored in the system and reflected in the passenger application.
Comments	This is a critical management use case that directly impacts trip availability and overall passenger experience.

Table A.5: Use Case Specification: Route and Vehicle Management

A.1.6 UC9: Safety, Support, and Reporting

Item	Description
Use Case Name	Safety, Support, and Reporting
Actors	Passenger: Reports safety issues, driver behavior, vehicle problems, and application issues; provides descriptions, attachments, and feedback; receives responses and status updates. Driver: Reports passenger misconduct, vehicle issues, safety concerns, or route problems; may provide evidence such as photos or audio. Admin: Handles technical issues, application-related problems, and safety emergencies; manages high-priority or sensitive reports; escalates or forwards reports when needed; logs resolutions and closes reports. Route Manager: Handles driver-related, route, and vehicle complaints; investigates reports; updates report status; coordinates corrective actions such as driver discipline or vehicle maintenance.
Description	This use case allows passengers and drivers to report safety concerns, misconduct, vehicle issues, or technical problems. The system automatically categorizes each report and assigns it to the appropriate responsible role to ensure timely handling and resolution.
Preconditions	User is logged into the system. The “Report Issue” interface is available.
Trigger	Passenger or driver selects “Report Issue” or “Safety & Support”.
Main Flow	1. User opens the “Report Issue / Safety” section. 2. The system displays issue categories, including driver-related issues, route or vehicle issues, application technical issues, and safety emergencies. 3. User selects the appropriate category. 4. User enters a description and may attach photos or audio files. 5. The system automatically assigns the report to the correct handler: Admin or Route Manager. 6. Admin or Route Manager reviews the report and updates its status. 7. The user receives a notification regarding the response or action taken.
Alternative Flow	A1 – Anonymous report: The user submits an anonymous safety complaint, which is delivered directly to the Admin. A2 – Driver reports a passenger: The Admin reviews the report and may forward it to the Route Manager if required.

Item	Description
Exception Flow	E1 – Required fields missing: The system prompts the user to complete missing fields. E2 – Weak internet connection: The report is saved offline and uploaded when connectivity is restored. E3 – File upload failure: The user may submit the report without attachments.
Data	Report ID, user ID, category, description, attachments, timestamps, assigned handler (Admin or Route Manager).
Postconditions	The report is created, stored, and assigned to the appropriate responsible party. Follow-up actions and resolutions are logged in the system.
Comments	Automatically routing reports to the correct role ensures faster issue resolution and aligns with defined system responsibilities.

Table A.6: Use Case Specification: Safety, Support, and Reporting

A.1.7 UC10: Map, GPS, and Navigation Services

Item	Description
Use Case Name	Map, GPS, and Navigation Services
Actors	Passenger: Views real-time driver location, routes, traffic, and ETA; interacts with the map (zoom, move, re-center); relies on GPS and map services for accurate tracking. Driver: Provides continuous GPS location updates; uses navigation mode for turn-by-turn directions; relies on ETA and route updates for efficient travel. GPS Service: Collects and delivers real-time geographic coordinates; detects movement, speed, and location accuracy; provides fallback methods such as WiFi-based location when GPS is weak. Map Service (Google Maps API): Renders maps, routes, traffic, and navigation visuals; computes route geometry, distance, and ETA; updates the map periodically using real-time GPS data.
Description	This use case provides real-time GPS-based location services, route visualization, vehicle tracking, distance calculation, traffic awareness, and ETA estimation to support accurate navigation and live trip tracking.
Preconditions	GPS is enabled on the user device. Map API services are available. User has granted location permission.
Trigger	User opens the map screen, starts a trip, or tracks a driver.
Main Flow	1. The system requests and verifies GPS permission. 2. The GPS service retrieves the current location. 3. The map service displays the map, route line, traffic conditions, and nearby vehicles. 4. Passenger views the driver's live location on the map. 5. The map updates every few seconds with new GPS coordinates. 6. The system calculates ETA based on traffic, distance, and route information.
Alternative Flow	A1 – Weak GPS signal: The system switches to a WiFi-based location method to maintain tracking accuracy.
Exception Flow	E1 – GPS disabled: The system prompts the user to enable GPS. E2 – Maps API failure: The system displays a cached map with a “Service unavailable” message. E3 – Location permission denied: Tracking and navigation functions are disabled.
Data	GPS coordinates, route ID, ETA data, traffic status, trip ID.

Item	Description
Postconditions	Map loads successfully, locations are updated in real time, and trip tracking is displayed accurately.
Comments	This use case directly supports live tracking functionality and ETA accuracy throughout the system.

Table A.7: Use Case Specification: Map, GPS, and Navigation Services

A.2 Glossary (or data dictionary)

This section defines key terms and abbreviations used throughout the Navigo – وصلني system report to ensure clarity and consistent understanding.

- **ETA (Estimated Time of Arrival):** The estimated time at which a vehicle is expected to arrive at the pickup or destination location.
- **OTP (One-Time Password):** A temporary verification code sent to a user's phone number to verify identity during authentication.